Looped_Transformer

2026 年 5 月 17 日

1 Looped Transformer

1.1 理论

Looped Transformer 所有层共享相同的权重 θ，每层的输入是前一层输出与初始输入的和：

$$h_0 = \text{input}$$
$$h_l = \text{TransformerBlock}(h_{l-1} + h_0|\theta) \quad \text{for } l = 1, 2, ..., L$$

1.1.1 截断损失函数

$$\text{Loss}(\theta) = \mathbb{E}_P \left[\frac{1}{b - b_0} \sum_{t=b_0}^{b} \frac{1}{k+1} \sum_{i=0}^{k} (Y_t(P^i|\theta) - f(x_{i+1}))^2 \right]$$

其中 - $\min_\theta$：这是我们在寻找一组最优的物理定律（模型参数 θ）。- $\mathbb{E}_P$：对所有生成的 Prompt 分布求期望。在统计物理中，这相当于求 “系综平均”，即我们在无数种不同的初始条件（数据集）下测试这个定律是否普适。 - $\frac{1}{b-b_0}\sum_{t=b_0}^{b}$：这是时间平均。$b$ 是总迭代次数，T 是窗口大小，$b_0 = \max(b - T, 0)$。我们只关心系统演化到 “稳态” 附近的这段时间内 (也就是 b_0 到 b 的表现)，前面瞬态的误差我们不计较。 - $\frac{1}{k+1}\sum_{i=0}^{k}$：这是空间/序列平均。也就是在给定不同长度的上下文 P^i 时，模型预测下一个点的值与真实函数 $f(x_{i+1})$ 之间的均方误差。 - $Y_t(P^i|\theta)$：在时间 t、给定历史信息 P^i 和系统参数 θ 时，系统当前的状态输出。

1.1.2 下游任务

- 数据生成方式：这篇论文解决的是纯粹的数学数据拟合问题。数据集是在训练循环中即时生成的浮点数向量。
- 以最基础的 **线性回归（Linear Regression）** 为例：
  - 维度设定：问题维度 $d = 20$，上下文样本数 $k = 40$。
  - 生成逻辑：每次迭代前，随机采样一个真实的参数 $w \sim \mathcal{N}(0, I_d)$。
  - 采样输入：生成 $k + 1$ 个输入样本（包含测试样本）$x_i \sim \mathcal{N}(0, I_d)$。

  - 计算标签：$y_i = w^T x_i$。
  - 拼接成 Prompt：最终喂给模型的序列 P 是一个交替的序列 $(x_1, y_1, x_2, y_2, \dots, x_k, y_k, x_{test}, y_{dummy})$。
- 线性回归可以进一步改成非线性回归（比如 ReLU 与 Linear 叠加），或者更复杂的物理系统。

把 Prompt 映射到高维 Embedding 空间　双通道映射与“拉链式”交织

由于 `x` 和 `y` 代表不同的物理量（一个是空间坐标/特征，一个是目标观测值），我们需要赋予它们独立的投影空间：

1. 定义两个独立的线性层：

```
read_in_x = nn.Linear(20, 256)
read_in_y = nn.Linear(1, 256)
```

2. 把前 `k` 个 `x_i` 丢给 `read_in_x`，变成张量 `X_emb`，形状为 `[batch, k, 256]`。
3. 把前 `k` 个 `y_i` 丢给 `read_in_y`，变成张量 `Y_emb`，形状为 `[batch, k, 256]`。
4. 像拉拉链一样，在 `seq_len` 维度上把 `X_emb`、`Y_emb` 交织（Interleave）起来，拼接成最终长度为 `2k` 的提示向量 `P`。

在工程实现中，我们在 x_{test} 后填充了一个 y_{dummy}（如 0），使实际拼接长度变为 `2k+2`（偶数）。因此，真正对 x_{test} 的预测结果位于输出序列的倒数第二个位置，即索引 `[-2]` 处。

1.2 代码

[101]:
```
import torch
import torch.nn as nn
import torch.nn.functional as F
import matplotlib.pyplot as plt
import time
from operator import sub
import copy
import numpy as np
import math
import concurrent.futures
import threading
```

1.2.1 位置编码 (PE, Position Encoding)

绝对位置编码 (APE, Absolute Position Encoding)

[102]:
```python
class APE(nn.Module):
    def __init__(self, d_model, max_seq_len=4096, b=10000):
        super().__init__()
        theta_i = 1 / (b ** (torch.arange(0, d_model, 2).float() / d_model)) # [d_model/2]
        m = torch.arange(max_seq_len).float() # [max_seq_len]
        m_theta_i = torch.outer(m, theta_i) # [max_seq_len, d_model/2]
        # APE 核心：直接生成一个完整的 pe 矩阵，偶数维度放 sin，奇数维度放 cos
        stacked=torch.stack((torch.sin(m_theta_i), torch.cos(m_theta_i)),dim=-1) # [max_seq_len, d_model/2, 2]
        pe=stacked.flatten(1,2) # [max_seq_len, d_model]
        self.register_buffer('pe', pe.unsqueeze(0)) # shape: [1, max_seq_len, d_model]

    def forward(self, x):
        # x shape: [batch_size, seq_len, d_model]
        return x + self.pe[:, :x.shape[1], :]
```

Learned APE

[103]:
```python
class LearnedAPE(nn.Module):
    def __init__(self, d_model, max_seq_len=4096):
        super().__init__()
        self.pe = nn.Embedding(max_seq_len, d_model)

    def forward(self, x):
        # x shape: [batch_size, seq_len, d_model]
        seq_len = x.shape[1]
        positions = torch.arange(seq_len, device=x.device)
        return x + self.pe(positions).unsqueeze(0)
```

ALiBi (Attention with Linear Biases) ALiBi 通过更改内积的方式引入线性位置偏置，从而实现了位置编码的功能。

$$\text{Score}(q_i, k_j) = q_i \cdot k_j - m \cdot (i - j) \quad (i \geq j)$$

其中 m 是一个与头数 (H) 相关的斜率参数，硬编码为

$$m_h = 2^{-\frac{8h}{H}} \quad h = 0, 1, ..., H-1$$

[104]:
```python
class ALiBi(nn.Module):
    def __init__(self, num_heads, max_seq_len=4096):
        super().__init__()
        dist = torch.arange(max_seq_len) # [max_seq_len]
        dist_matrix = torch.abs(dist.view(-1, 1) - dist.view(1, -1))  # [max_seq_len, max_seq_len]
        bias = -(2**-(8*torch.arange(1,num_heads+1) / num_heads)).view(-1, 1, 1) * dist_matrix.unsqueeze(0)  # [num_heads, max_seq_len, max_seq_len]
        # 因果掩码
        tril = torch.tril(torch.ones(max_seq_len, max_seq_len)).bool() # torch.tril (triangular lower) 提取下三角元素，并把上三角元素置 0
        mask = torch.zeros(max_seq_len, max_seq_len).masked_fill(~tril, float('-inf')) # 下三角元素置 0，上三角元素置-inf
        self.register_buffer('fused_mask', bias + mask) # [num_heads, max_seq_len, max_seq_len]

    def forward(self, seq_len):
        # score shape: [batch_size, num_heads, seq_len, seq_len]
        return self.fused_mask[:, :seq_len, :seq_len]
```

原来的 RoPE (Rotary Position Encoding)

[105]:
```python
class RoPE(nn.Module):
    def __init__(self, d_k, max_seq_len = 4096, b=10000):
        super().__init__()
        theta_i = 1/(b**(torch.arange(0,d_k,2).float()/d_k))  # \theta_i = b^{-\frac{2i}{d}}, \quad i \in \{0,1,\dots,\frac{d}{2}-1\}
        m = torch.arange(max_seq_len).float()  # m = [0,1,...,seq_len-1]
        m_theta_i = torch.outer(m, theta_i)  # [seq_len, d_k/2]
        cos = torch.cos(torch.cat((m_theta_i, m_theta_i), dim=-1))  # [seq_len, d_k]
        sin = torch.sin(torch.cat((m_theta_i, m_theta_i), dim=-1))  # [seq_len, d_k]
        self.register_buffer('cos', cos[None, None, :, :])  # 好处：cos 和 sin 不需要更新参数，注册为 buffer 后会自动放到正确的设备上
        self.register_buffer('sin', sin[None, None, :, :])  # 扩充维度以适应后续计算
```

```
    def forward(self, x): # 适用于 Q、K (V 不需要位置编码!)
        # x: [batch_size, num_heads, seq_len, d_k]
        seq_len = x.shape[2]
        d_2 = x.shape[-1] // 2
        cos = self.cos[:, :, :seq_len, :]  # [1, 1, seq_len, d_k]
        sin = self.sin[:, :, :seq_len, :]
        x_first_half = x[..., :d_2]
        x_second_half = x[..., d_2:]
        x_flip = torch.cat((-x_second_half, x_first_half), dim=-1)
        x_out = x * cos + x_flip * sin  # 旋转位置编码
        return x_out
```

自创：Multi-Scale Untied Positional Encoding (MS-UPE)

[106]:
```
class MS_UPE(nn.Module):
    def __init__(self,num_heads, d_k, max_seq_len = 4096, b_0=10000,␣
↪head_ratio=2):
        super().__init__()
        b=(b_0*(head_ratio**(torch.arange(0, num_heads)))).float() # [num_heads]
        # \theta_i = b^{-\frac{2i}{d}}, \quad i \in \{0,1,\dots,\frac{d}{2}-1\}
        theta_i = 1/(b.unsqueeze(1)**(torch.arange(0,d_k,2).float()/d_k).
↪unsqueeze(0))  # [num_heads, d_k/2]
        m = torch.arange(max_seq_len).float()  # m = [0,1,...,seq_len-1]
        m_theta_i = torch.einsum('m,hk->hmk', m, theta_i)  # [num_heads,␣
↪seq_len, d_k/2]
        pe = torch.cat((torch.cos(m_theta_i), torch.sin(m_theta_i)), dim=-1)  #␣
↪[num_heads, seq_len, d_k]
        self.register_buffer('pe', pe[None, :, :, :])  # 好处: pe 不需要更新参数,
注册为 buffer 后会自动放到正确的设备上

    def forward(self, x): # 适用于 Q、K (V 不需要位置编码!)
        # x: [batch_size, num_heads, seq_len, d_k]
        seq_len = x.shape[2]
        pe = self.pe[:, :, :seq_len, :]  # [1, num_heads, seq_len, d_k]
        x_out = x + pe
        return x_out
```

1.2.2 Multi-Head Attention

注：APE 应该加在 ToyModel 上而不是 TransformerBlock 上，因为我们只需要最开头输入处编码位置，而不是叠加 num_blocks 次。

[107]:
```
class MultiHeadAttention(nn.Module):
    def __init__(self, num_heads, d_model, max_seq_len=4096,␣
↪pe_type='learned_ape', b_rope_or_upe=10000, head_ratio_upe=2):
        super().__init__()
        assert d_model % num_heads == 0, "d_model must be divisible by␣
↪num_heads"
        # 维度属性
        self.num_heads = num_heads
        self.d_model = d_model
        self.d_k = d_model // num_heads
        # 权重矩阵
        self.W_Q = nn.Linear(d_model, d_model, bias=False)
        self.W_K = nn.Linear(d_model, d_model, bias=False)
        self.W_V = nn.Linear(d_model, d_model, bias=False)
        self.W_O = nn.Linear(d_model, d_model, bias=False)
        # 位置编码模块
        self.pe_type = [pe_type] if isinstance(pe_type, str) else pe_type
        self.qk_pe_modules = nn.ModuleList()
        self.sc_pe_modules = nn.ModuleList()
        for pe in self.pe_type:
            if pe == 'rope':
                pe_module = RoPE(self.d_k, max_seq_len, b=b_rope_or_upe)
                self.qk_pe_modules.append(pe_module)  # RoPE 应用于 Q 和 K
            elif pe == 'ms_upe':
                pe_module = MS_UPE(num_heads, self.d_k, max_seq_len,␣
↪b_0=b_rope_or_upe, head_ratio=head_ratio_upe)
                self.qk_pe_modules.append(pe_module)  # MS-UPE 应用于 Q 和 K
            elif pe == 'alibi':
                pe_module = ALiBi(num_heads, max_seq_len)
                self.sc_pe_modules.append(pe_module)  # ALiBi 应用于 score 矩阵
            elif pe in ['ape', 'learned_ape']:
                pass  # APE 和 Learned APE 在输入时直接加到 x 上，不需要单独模块
```

```
        # 因果掩码
        tril = torch.tril(torch.ones(max_seq_len, max_seq_len)).bool() # torch.
↪tril (triangular lower) 提取下三角元素，并把上三角元素置 0
        mask = torch.zeros(max_seq_len, max_seq_len).masked_fill(~tril,␣
↪float('-inf')) # 下三角元素置 0，上三角元素置-inf
        self.register_buffer('mask', mask[None, None, :, :])  # [1, 1, seq_len,␣
↪seq_len]

    def forward(self, x): # x: [batch_size, seq_len, d_model]
        # 线性投影与分头
        batch_size, seq_len, _ = x.shape
        Q = self.W_Q(x).view(batch_size, seq_len, self.num_heads, self.d_k).
↪transpose(1, 2)  # [batch_size, num_heads, seq_len, d_k]
        K = self.W_K(x).view(batch_size, seq_len, self.num_heads, self.d_k).
↪transpose(1, 2)  # 交换 (1, 2) 是因为之后 softmax 时要进行矩阵乘法 (只乘后两维)
        V = self.W_V(x).view(batch_size, seq_len, self.num_heads, self.d_k).
↪transpose(1, 2)  # 在 seq_len 维度上分配注意力
        # 根据选择的 PE 类型对 Q 和 K 进行位置编码
        for pe_module in self.qk_pe_modules:
            Q = pe_module(Q)
            K = pe_module(K)
        if not self.training:
            # 缩放点积注意力与因果掩码 Causal Masking
            ## Query-Key 点积计算注意力得分
            scores = torch.matmul(Q, K.transpose(-2, -1)) / (self.d_k ** 0.5) ␣
↪# [batch_size, num_heads, seq_len, seq_len]
            if self.sc_pe_modules:
                fused_mask = self.sc_pe_modules[0](seq_len)  # [num_heads,␣
↪seq_len, seq_len]
                scores = scores + fused_mask  # 将 ALiBi 的偏置加到得分上
                attention = F.softmax(scores, dim=-1)  # [batch_size,␣
↪num_heads, seq_len, seq_len]
            else:
                ## 因果掩码和 softmax 计算注意力权重
```

```
                attention = F.softmax(scores + self.mask[..., :seq_len, :
↪seq_len], dim=-1)  # [batch_size, num_heads, seq_len, seq_len]
            self.captured_attention = attention  # ** 新增 **: 捕获注意力矩阵，用
hook 记录下来，方便后续分析
            ## 乘以 Value 完成加权求和
            out = torch.matmul(attention, V)  # [batch_size, num_heads,␣
↪seq_len, d_k]
        else:
            if self.sc_pe_modules:
                fused_mask = self.sc_pe_modules[0](seq_len)  # [num_heads,␣
↪seq_len, seq_len]
                out = F.scaled_dot_product_attention(Q, K, V,␣
↪attn_mask=fused_mask, is_causal=False)  # [batch_size, num_heads, seq_len,␣
↪d_k]
            else:
                out = F.scaled_dot_product_attention(Q, K, V, is_causal=True) ␣
↪# [batch_size, num_heads, seq_len, d_k]
        # 多头拼接并乘以输出矩阵 (必须先内存连续化 (contiguous) 再 view，否则会报
错)
        out = out.transpose(1, 2).contiguous().view(batch_size, seq_len, -1)  #␣
↪[batch_size, seq_len, d_model]
        H = self.W_O(out)  # [batch_size, seq_len, d_model]
        return H
```

1.2.3 SwiGLU 和单个 Transformer Block 架构不变

[108]:
```
class SwiGLU(nn.Module):
    def __init__(self,d_model,d_hidden):
        super().__init__()
        self.W = nn.Linear(d_model,d_hidden,bias=False)
        self.V = nn.Linear(d_model,d_hidden,bias=False)
        self.W2 = nn.Linear(d_hidden,d_model,bias=False)
    def forward(self,x):
        x1 = F.silu(self.W(x))
        x2 = self.V(x)
        x_out = self.W2(x1 * x2)
```

```
        return x_out
```

[109]:
```
class TransformerBlock(nn.Module):
    def __init__(self, num_heads, d_model, max_seq_len=4096, multiplier=4,␣
↪norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',␣
↪b_rope_or_upe=10000, head_ratio_upe=2):
        super().__init__()
        if norm_type == 'rmsnorm':
            self.norm1 = nn.RMSNorm(d_model)  # 第一次归一化
            self.norm2 = nn.RMSNorm(d_model)  # 第二次归一化
        elif norm_type == 'layernorm':
            self.norm1 = nn.LayerNorm(d_model)  # 第一次归一化
            self.norm2 = nn.LayerNorm(d_model)  # 第二次归一化
        self.attention = MultiHeadAttention(num_heads, d_model,␣
↪max_seq_len=max_seq_len, pe_type=pe_type, b_rope_or_upe=b_rope_or_upe,␣
↪head_ratio_upe=head_ratio_upe)
        if ffn_type == 'swiglu':
            self.ffn = SwiGLU(d_model, d_model * multiplier)
        elif ffn_type == 'gelu':
            self.ffn = nn.Sequential(
                nn.Linear(d_model, d_model * multiplier),
                nn.GELU(),
                nn.Linear(d_model * multiplier, d_model)
            )

    def forward(self, x):
        # 第一次归一化 Pre-Norm
        x_norm1 = self.norm1(x)
        # 全局信息交互 Multi-Head Attention (MHA)
        h = self.attention(x_norm1)
        # 第一次残差连接
        x1 = x + h
        # 第二次归一化
        x_norm2 = self.norm2(x1)
        # 前馈网络 Feed-Forward Network (FFN)
        x_out = self.ffn(x_norm2)
```

```
        # 第二次残差连接
        x2 = x1 + x_out
        return x2
```

1.2.4 保留 Attention Probe

[110]:
```
class AttentionProbe:
    def __init__(self):
        self.captured_data = []
    def __call__(self, module, input, output):
        if getattr(module, 'captured_attention', None) is not None:
            attention = module.captured_attention.detach().cpu().numpy()
            self.captured_data.append(attention)
        else:
            self.captured_data.append(None)  # 如果没有捕获到注意力矩阵，记录一个
None 占位符
    def reset(self):
        self.captured_data = []
```

1.2.5 Toy Model: Looped Transformer

[111]:
```
class ToyModel(nn.Module):
    def __init__(self, num_blocks, num_heads, d_model, max_seq_len=4096,
                 norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',␣
↪b_rope_or_upe=10000, head_ratio_upe=2,
                 loop=True, residual_gate=(1,1), residual_gate_type='fixed',␣
↪residual_random=(1,0.1)):
        super().__init__()
        self.loop = loop
        if loop:
            self.transformer_block = TransformerBlock(num_heads, d_model,␣
↪max_seq_len=max_seq_len, norm_type=norm_type, ffn_type=ffn_type,␣
↪pe_type=pe_type, b_rope_or_upe=b_rope_or_upe, head_ratio_upe=head_ratio_upe)␣
↪# 这里保证了各层权重始终相同
            self.probe = AttentionProbe()
            self.transformer_block.attention.register_forward_hook(self.probe)
            self.captured_attention = None
```

```
        else:
            self.transformer_block = nn.ModuleList([
                TransformerBlock(num_heads, d_model, max_seq_len=max_seq_len,␣
↪norm_type=norm_type, ffn_type=ffn_type, pe_type=pe_type,␣
↪b_rope_or_upe=b_rope_or_upe, head_ratio_upe=head_ratio_upe)
                for _ in range(num_blocks)
            ])
            self.probe = [AttentionProbe() for _ in range(num_blocks)]
            for block, probe in zip(self.transformer_block, self.probe):
                block.attention.register_forward_hook(probe)

        residual_random = torch.tensor(residual_random, dtype=torch.float32)
        if residual_gate == 'random':
            residual_gate_tensor = torch.
↪randn(2)*residual_random[1]+residual_random[0] # 随机初始化门控参数，初始正态分布
        else:
            residual_gate_tensor = torch.tensor(residual_gate, dtype=torch.
↪float32)  # 将输入的元组转换为张量，方便后续计算
        if residual_gate_type == 'fixed':
            self.register_buffer('residual_gate', residual_gate_tensor)  # 固定的门控参数，注册为 buffer 使其成为模型状态的一部分，但不参与梯度更新
        elif residual_gate_type == 'learnable_scalar':
            self.residual_gate = nn.Parameter(torch.
↪ones(2)*residual_gate_tensor)  # 可学习的标量门控参数，初始值为 1.0
        elif residual_gate_type == 'learnable_vector':
            if residual_gate == 'random':
                self.residual_gate = nn.Parameter(torch.randn(d_model,␣
↪2)*residual_random[1]+residual_random[0]) # 可学习的向量门控参数，初始值为随机向量
            else:
                self.residual_gate = nn.Parameter(torch.ones(d_model,␣
↪2)*residual_gate_tensor)  # 可学习的向量门控参数，初始值为全 1 向量

        pe_type = self.transformer_block.attention.pe_type
        self.ape = nn.ModuleList()
```

```
        for pe in pe_type:
            if pe == 'ape':
                self.ape.append(APE(d_model, max_seq_len=max_seq_len))
            elif pe == 'learned_ape':
                self.ape.append(LearnedAPE(d_model, max_seq_len=max_seq_len))
            elif pe in ['rope', 'ms_upe']:
                pass # RoPE 和 MS_UPE 在 MultiHeadAttention 内部处理位置编码，不需要单独的模块在这里处理

        self.num_blocks = num_blocks
        self.num_heads = num_heads
        self.d_model = d_model
        self.max_seq_len = max_seq_len
        self.pe_type = pe_type

    def forward(self, x_0, num_eff=15, current_blocks=None):
        # num_eff 是需要加入误差计算的有效层数，也即理论公式中的 T
        # x 的形状为 [batch_size, seq_len, d_model]
        a=self.residual_gate[...,0]
        b=self.residual_gate[...,1]
        for ape_module in self.ape:
            x_0 = ape_module(x_0) # 绝对位置编码只加在最开始输入处
        x = x_0
        if current_blocks is None:
            current_blocks = self.num_blocks
        b_0 = max(0, current_blocks - num_eff) # 计算需要加入误差计算的有效层数对应的起始层索引
        outputs = []
        if self.loop:
            self.probe.reset() # 每次前向传播前重置捕获的数据，避免混淆不同次前向传播的注意力矩阵
            for i in range(current_blocks):
                if i == b_0:
                    x= x.detach() # 释放前面的计算图
                    x.requires_grad_(True)
                x = self.transformer_block(a*x+b*x_0)
```

```
                if i >= b_0:
                    outputs.append(x)
            self.captured_attention = list(self.probe.captured_data) # 显式 copy
一份数据，避免后续被修改
        else:
            for probe in self.probe:
                probe.reset()
            for i, block in enumerate(self.transformer_block):
                if i == b_0:
                    x= x.detach() # 释放前面的计算图
                    x.requires_grad_(True)
                x = block(a*x+b*x_0) # x: [batch_size, seq_len, d_model]
                if i >= b_0:
                    outputs.append(x)
            self.captured_attention = [probe.captured_data for probe in self.
probe]
        outputs = torch.stack(outputs, dim=1) # [batch_size, num_eff, seq_len,
d_model]
        return outputs
```

1.2.6 Downstream Task: Linear Regression

转换头

[112]:
```
class RegressionHead(nn.Module):
    def __init__(self, d_model, d_x, d_y=1, bias=False, init_scale=None):
        super().__init__()
        self.read_in_x = nn.Linear(d_x, d_model, bias=bias)
        self.read_in_y = nn.Linear(d_y, d_model, bias=bias)
        if init_scale is not None:
            nn.init.normal_(self.read_in_x.weight, mean=0.0, std=init_scale)
            nn.init.normal_(self.read_in_y.weight, mean=0.0, std=init_scale)
    def forward(self, x_data, y_data):
        # k 相当于 seq_len 的一半，因为 x 和 y 交织在一起，所以总的 seq_len 是 2k
        # x_data: [batch_size, k, d_x]
        # y_data: [batch_size, k, d_y]
        x_emb = self.read_in_x(x_data)  # [batch_size, k, d_model]
        y_emb = self.read_in_y(y_data)  # [batch_size, k, d_model]
```

```
        # 交织成 (x_1, y_1, x_2, y_2, ..., x_k, y_k)
        stacked = torch.stack((x_emb, y_emb), dim=2)  # [batch_size, k, 2,␣
↪d_model]
        Prompt = stacked.flatten(1,2)  # [batch_size, 2k, d_model]
        return Prompt
```

损失函数

[113]:
```
class PredictionLoss(nn.Module):
    def __init__(self, d_model, d_y=1, loss_type='mse', layer_weight_decay=1.0,␣
↪seq_weight_decay=1.0):
        super().__init__()
        self.read_out = nn.Linear(d_model, d_y)
        self.layer_weight_decay = layer_weight_decay
        self.seq_weight_decay = seq_weight_decay
        if loss_type == 'mse':
            self.loss_fn = nn.MSELoss(reduction='none')
        elif loss_type == 'l1':
            self.loss_fn = nn.L1Loss(reduction='none')

    def forward(self, outputs, y_true, is_eval=False, sink_padding=None):
        # outputs:[batch_size, num_eff, seq_len, d_model]
        # y_true: [batch_size, seq_len/2, 1]
        if is_eval:
            y_outputs = outputs[:, -1, -2, :] # 只取最后一层的最后一个位置的输出
作为预测值
            y_pred_final = self.read_out(y_outputs) # [batch_size, d_y]
            return y_pred_final
        y_outputs = outputs[:, :, 0::2, :] # [batch_size, num_eff, seq_len/2,␣
↪d_model]
        y_preds = self.read_out(y_outputs) # [batch_size, num_eff, seq_len/2,␣
↪d_y]
        if sink_padding is not None:
            y_preds = y_preds[:, :, sink_padding:, :]
            y_true = y_true[:, sink_padding:, :]
        y_pred_norm = torch.sqrt(torch.mean(y_preds.detach() ** 2)).item()
        y_true_norm = torch.sqrt(torch.mean(y_true.detach() ** 2)).item()
```

```
        loss_unreduced = self.loss_fn(y_preds, y_true.unsqueeze(1).
↪expand_as(y_preds)) # [batch_size, num_eff, seq_len/2, d_y]
        if self.layer_weight_decay != 1.0:
            num_eff = outputs.shape[1]
            layer_weights = self.layer_weight_decay ** torch.arange(num_eff-1,␣
↪-1, -1, device=outputs.device)  # 从最后一层到第一层递减的权重
            layer_weights = layer_weights / layer_weights.mean()
            loss_unreduced = loss_unreduced * layer_weights.view(1, num_eff, 1,␣
↪1)  # 按层加权
        if self.seq_weight_decay != 1.0:
            k=loss_unreduced.shape[2]
            seq_weights = self.seq_weight_decay ** torch.arange(k-1, -1, -1,␣
↪device=outputs.device)  # 从最后一次预测到第一次预测递减的权重
            seq_weights = seq_weights / seq_weights.mean()
            loss_unreduced = loss_unreduced * seq_weights.view(1, 1, k, 1)  # 按
位置加权
        return loss_unreduced.mean(), y_pred_norm, y_true_norm
```

拼装

[114]:
```
class RegressionSolver(nn.Module):
    def __init__(self, num_blocks, num_heads, d_model, d_x, d_y, max_seq_len,
                 norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',␣
↪b_rope_or_upe=10000, head_ratio_upe=2,
                 loop=True, loss_type='mse',
                 residual_gate=(1,1), residual_gate_type='fixed',␣
↪residual_random=(1,0.1),
                 bias=False, init_scale=None,
                 layer_weight_decay=1.0, seq_weight_decay=1.0):
        super().__init__()
        self.toy_model = ToyModel(num_blocks, num_heads, d_model, max_seq_len,
                                  norm_type=norm_type, ffn_type=ffn_type,␣
↪pe_type=pe_type,
                                  loop=loop, b_rope_or_upe=b_rope_or_upe,␣
↪head_ratio_upe=head_ratio_upe,
                                  residual_gate=residual_gate,␣
↪residual_gate_type=residual_gate_type, residual_random=residual_random)
```

```
        self.head = RegressionHead(d_model, d_x, d_y, bias=bias,␣
↪init_scale=init_scale)
        self.loss_fn = PredictionLoss(d_model, d_y, loss_type=loss_type,␣
↪layer_weight_decay=layer_weight_decay, seq_weight_decay=seq_weight_decay)

    def forward(self, x_data, y_data, num_eff, current_blocks=None,␣
↪is_eval=False, sink_padding=None):
        Prompt = self.head(x_data, y_data) # [batch_size, seq_len, d_model]
        outputs = self.toy_model(Prompt, num_eff,␣
↪current_blocks=current_blocks) # [batch_size, num_eff, seq_len, d_model]
        return self.loss_fn(outputs, y_data, is_eval=is_eval,␣
↪sink_padding=sink_padding)
```

1.2.7 数据生成与加载

数据生成

线性回归数据

[115]:
```
def linear_data_generator(batch_size, seq_len, d_x, d_y, device='cpu',␣
↪generator=None):
    w = torch.randn(batch_size, d_x, d_y, device=device, generator=generator)/␣
↪(d_x ** 0.5)  # 真实线性函数 w: [batch_size, d_x, d_y]
    x_data = torch.randn(batch_size, seq_len//2, d_x, device=device,␣
↪generator=generator)  # 输入特征 x: [batch_size, seq_len//2, d_x]
    y_data = x_data @ w  # 计算标签 y = x @ w: [batch_size, seq_len//2, d_y]
    return x_data, y_data
```

非线性回归数据

[116]:
```
def nonlinear_data_generator(batch_size, seq_len, d_x, d_y, d_hidden,␣
↪function_callable, device='cpu', generator=None, nonlinear_funcs=['relu']):
    '''
    linear(d_x->d_hidden) + nonlinear_func + linear(d_hidden->d_y) 的形式生成数
据
    function_callable: 非线性函数，如 lambda x: F.relu(2*torch.sin(x))+torch.
↪cos(x)
    d_hidden: 隐藏层维度，控制非线性函数的输入输出维度。
```

```
        若 d_hidden=d_x，则第一层线性变换默认设定为恒等映射，数据生成过程变为 nonlinear_func + linear 的形式；
        若 d_hidden=d_y，则第二层线性变换默认设定为恒等映射，数据生成过程变为 linear + nonlinear_func 的形式。
    '''
    w1 = torch.randn(batch_size, d_x, d_hidden, device=device,␣
↪generator=generator)/ (d_x ** 0.5)  # 第一层权重 w1: [batch_size, d_x,␣
↪d_hidden]
    w2 = torch.randn(batch_size, d_hidden, d_y, device=device,␣
↪generator=generator)/ (d_hidden ** 0.5)  # 第二层权重 w2: [batch_size,␣
↪d_hidden, d_y]
    x_data = torch.randn(batch_size, seq_len//2, d_x, device=device,␣
↪generator=generator)  # 输入特征 x: [batch_size, seq_len//2, d_x]
    if d_x==d_hidden:
        w1=torch.eye(d_x, device=device).unsqueeze(0).expand(batch_size, -1,␣
↪-1) # 如果输入维度和隐藏层维度相同，变为 nonlinear_func + linear 的形式
    if d_hidden==d_y:
        w2=torch.eye(d_hidden, device=device).unsqueeze(0).expand(batch_size,␣
↪-1, -1) # 如果隐藏层维度和输出维度相同，变为 linear + nonlinear_func 的形式
    hidden = x_data @ w1  # [batch_size, seq_len//2, d_hidden]
    # 应用非线性函数
    hidden = function_callable(hidden)  # [batch_size, seq_len//2, d_hidden]
    # 第二层线性变换
    y_data = hidden @ w2  # [batch_size, seq_len//2, d_y]
    return x_data, y_data
```

数据加载器

[117]:
```
def dataloader(batch_size=64, seq_len=80, d_x=20, d_y=1, device='cpu',␣
↪data_type='linear', d_hidden=None, function_callable=None, sink_padding=1,␣
↪generator=None):
    while True:
        if data_type == 'linear':
            x_data, y_data = linear_data_generator(batch_size, seq_len, d_x,␣
↪d_y, device=device, generator=generator)
        elif data_type == 'nonlinear':
```

```
            x_data, y_data = nonlinear_data_generator(batch_size, seq_len, d_x,␣
↪d_y, d_hidden, function_callable, device=device, generator=generator)
        if sink_padding is not None:
            x_data = F.pad(x_data, (0, 0, sink_padding, 0), value=0)  # 在
seq_len 维度最前面添加 sink_padding 个全 0 向量
            y_data = F.pad(y_data, (0, 0, sink_padding, 0), value=0)
        yield x_data, y_data
```

1.2.8 实验模块

[118]:
```
class LoopedTransformerExperiment:
    def __init__(self, num_blocks, num_heads=8, d_model=256, lr=1e-4,␣
↪gate_lr_ratio=100, max_seq_len=4096, d_x=20, d_y=1,
                 seed=None, experiment_name='Experiment',
                 norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',␣
↪b_rope_or_upe=10000, head_ratio_upe=2,
                 optimizer_type='adam',wd_adamw=0.01,
                 loop=True, loss_type='mse', bias=False, init_scale=None,
                 residual_gate=(1,1), residual_gate_type='fixed',␣
↪residual_random=(1,0.1),
                 task='regression', timing=True,
                 layer_weight_decay=1.0, seq_weight_decay=1.0):
        if timing:
            start_time = time.time()
        # 设备选择：优先使用 MPS（适用于 Apple Silicon），其次是 CUDA（适用于␣
↪NVIDIA GPU），最后退回 CPU
        if torch.backends.mps.is_available():
            self.device = torch.device("mps")
        elif torch.cuda.is_available():
            self.device = torch.device("cuda")
        else:
            self.device = torch.device("cpu")
        self.is_offloaded = False
        self.generator = torch.Generator(device=self.device)
        if seed is not None:
            torch.manual_seed(seed)
```

```
            self.generator.manual_seed(seed)
        # 模型和优化器
        self.task = task
        if task == 'regression':
            self.model = RegressionSolver(num_blocks, num_heads, d_model,␣
↪d_x=d_x, d_y=d_y, max_seq_len=max_seq_len,
                                          norm_type=norm_type,␣
↪ffn_type=ffn_type, pe_type=pe_type, b_rope_or_upe=b_rope_or_upe,
                                          head_ratio_upe=head_ratio_upe,␣
↪loop=loop, loss_type=loss_type,
                                          bias=bias, init_scale=init_scale,
                                          residual_gate=residual_gate,␣
↪residual_gate_type=residual_gate_type, residual_random=residual_random,
                                         ␣
↪layer_weight_decay=layer_weight_decay, seq_weight_decay=seq_weight_decay
                                          ).to(self.device)
        gate_params=[]
        base_params=[]
        for name, param in self.model.named_parameters():
            if 'residual_gate' in name:
                gate_params.append(param)
            else:
                base_params.append(param)
        param_groups = [{'params': base_params},
                        {'params': gate_params, 'lr': lr * gate_lr_ratio}]
        if optimizer_type == 'adam':
            self.optimizer = torch.optim.Adam(param_groups, lr=lr)
        elif optimizer_type == 'sgd':
            self.optimizer = torch.optim.SGD(param_groups, lr=lr)
        elif optimizer_type == 'adamw':
            self.optimizer = torch.optim.AdamW(param_groups, lr=lr,␣
↪weight_decay=wd_adamw)
        self.d_x = d_x
        self.d_y = d_y
        self.num_blocks = num_blocks
        self.init_time = None
```

```
        if experiment_name is not None:
            if timing:
                self.init_time = time.time() - start_time
                print(f"{experiment_name} initialized in {self.init_time:.2f}␣
↪seconds. Using device: {self.device}")
            else:
                print(f"{experiment_name} initialized. Using device: {self.
↪device}")

    def train(self, batch_size=64, seq_len=80, epochs=20, data_type='linear',
              d_hidden=None, function_callable=None,
              num_eff=15, sink_padding=None, scheduled_training=True,␣
↪scheduler_type=None, lr_scale=0.1, step_size_scheduler=10,
              print_every=None, timing=False):
        # x_data: [batch_size, k, d_x]
        # y_data: [batch_size, k, d_y]
        self.model.train()
        self.max_torch_vram = 0
        self.max_metal_vram = 0
        self.train_time = None
        self.data_loader = dataloader(batch_size, seq_len, self.d_x, self.d_y,␣
↪device=self.device, data_type=data_type, sink_padding=sink_padding,␣
↪generator=self.generator, d_hidden=d_hidden,␣
↪function_callable=function_callable)
        self.loss_history = []
        self.y_pred_norm_history = []
        self.y_true_norm_history = []
        self.residual_gate_history = []
        if scheduler_type == 'cosine':
            scheduler = torch.optim.lr_scheduler.CosineAnnealingLR(self.
↪optimizer, T_max=epochs, eta_min=lr_scale*self.optimizer.
↪param_groups[0]['lr'])
        elif scheduler_type == 'step':
            scheduler = torch.optim.lr_scheduler.StepLR(self.optimizer,␣
↪step_size=step_size_scheduler, gamma=lr_scale)
        num_eff = min(num_eff, self.num_blocks) # 确保 num_eff 不超过总层数
```

```
        if timing:
            start_time = time.time()
        for epoch in range(epochs):
            self.optimizer.zero_grad()
            x_data, y_data = next(self.data_loader) # 从数据加载器中获取一个批次
的数据
            if scheduled_training:
                current_blocks = min(num_eff + epoch, self.num_blocks) # 随着训
练的进行，逐渐增加参与误差计算的有效层数
            else:
                current_blocks = self.num_blocks
            loss, y_pred_norm, y_true_norm = self.model(x_data, y_data,␣
↪num_eff=num_eff, current_blocks=current_blocks, is_eval=False,␣
↪sink_padding=sink_padding) # 前向传播计算损失 [batch_size, num_eff, seq_len/
↪2, d_y]
            loss = loss.mean()  # 对所有维度求平均得到一个标量损失值
            loss.backward()  # 反向传播计算梯度
            if torch.backends.mps.is_available():
                self.max_metal_vram = max(self.max_metal_vram, torch.mps.
↪driver_allocated_memory() / (1024 ** 3))
                self.max_torch_vram = max(self.max_torch_vram, torch.mps.
↪current_allocated_memory() / (1024 ** 3))
            self.optimizer.step()  # 更新模型参数
            if torch.backends.mps.is_available():
                torch.mps.empty_cache()  # 清空 MPS 缓存，释放未使用的显存
            elif torch.cuda.is_available():
                torch.cuda.empty_cache()  # 清空 CUDA 缓存，释放未使用的显存
            if scheduler_type is not None:
                scheduler.step()  # 更新学习率
            self.y_pred_norm_history.append(y_pred_norm)
            self.y_true_norm_history.append(y_true_norm)
            self.loss_history.append(loss.item())  # 记录损失值
            self.residual_gate_history.append(self.model.toy_model.
↪residual_gate.detach().cpu().numpy() if hasattr(self.model.toy_model,␣
↪'residual_gate') else None)  # 记录门控参数值
            if print_every is not None and (epoch + 1) % print_every == 0:
```

```
                print(f"Epoch {epoch+1}/{epochs}, Loss: {loss.item():.4f}, Norm␣
↪Ratio: {y_pred_norm/y_true_norm if y_true_norm != 0 else None:.4f}, Current␣
↪Blocks: {current_blocks}")
        if timing:
            self.train_time = time.time()-start_time
            print(f"Training completed in {self.train_time:.2f} seconds.")
        self.residual_gate_values = self.residual_gate_history[-1]
        self.final_loss = self.loss_history[-1]

    def evaluate(self, batch_size=64, seq_len=80, data_type='linear',␣
↪loss_type='mse', sink_padding=1):
        self.model.eval()
        with torch.no_grad():
            # 在评估模式下，直接使用所有层的输出进行预测（因为已经 no_grad 了，所以
不需要防止梯度爆炸了）
            x_eval, y_eval = next(dataloader(batch_size, seq_len+1, self.d_x,␣
↪self.d_y, device=self.device, data_type=data_type,␣
↪sink_padding=sink_padding, generator=self.generator, d_hidden=None,␣
↪function_callable=None)) # 生成一个评估用的数据批次，注意这里的 seq_len 要比训练
时大 1，因为我们需要预测最后一个位置的 y 值
            y_pred = self.model(x_eval, y_eval, num_eff=self.num_blocks,␣
↪current_blocks=self.num_blocks, is_eval=True, sink_padding=sink_padding) #␣
↪[batch_size, d_y]
            if loss_type == 'mse':
                mse=F.mse_loss(y_pred, y_eval[:, -1, :])
                print(f"MSELoss: {mse:.4f}") # 计算预测值与真实值之间的均方误差
                return mse.item()
            elif loss_type == 'l1':
                l1=F.l1_loss(y_pred, y_eval[:, -1, :])
                print(f"L1Loss: {l1:.4f}")
                return l1.item()

    def get_results(self, result_list:list[str]):
        all_results = {
            'init_time': self.init_time,                                    #␣
↪float: 实验初始化时间（秒）
```

```
            'train_time': self.train_time,                                  # float: 训练时间（秒）
            'loss_history': self.loss_history,                              # list(float): 训练过程中每个 epoch 的损失值
            'y_pred_norm_history': self.y_pred_norm_history,                # list(float): 训练过程中每个 epoch 的预测值范数
            'y_true_norm_history': self.y_true_norm_history,                # list(float): 训练过程中每个 epoch 的真实值范数
            'y_norm_error_history': [pred - true for pred, true in zip(self.y_pred_norm_history, self.y_true_norm_history)], # list(float): 训练过程中每个 epoch 的预测值范数与真实值范数之差的绝对值
            'y_norm_ratio_history': [pred / true if true != 0 else None for pred, true in zip(self.y_pred_norm_history, self.y_true_norm_history)], # list(float): 训练过程中每个 epoch 的预测值范数与真实值范数之比
            'final_loss': self.final_loss,                                  # float: 最终的损失值
            'final_y_pred_norm': self.y_pred_norm_history[-1],              # float: 最终的预测值范数
            'final_y_true_norm': self.y_true_norm_history[-1],              # float: 最终的真实值范数
            'residual_gate_history': self.residual_gate_history,            # list[np.array[2] 或 np.array[d_model, 2]]: 训练过程中每个 epoch 的残差门控参数值
            'final_residual_gate': self.residual_gate_values,               # np.array[2] 或 np.array[d_model, 2]: 最终的残差门控参数值
            'captured_attention': self.model.toy_model.captured_attention # list[np.array[batch_size, num_heads, seq_len, seq_len]]: 捕获的注意力权重
            }
        if any("residual_gate" in key for key in result_list) and self.residual_gate_values is not None:
            if len(self.residual_gate_values.shape) == 1:
                all_results['residual_gate_history_a'] = [gate[0] for gate in self.residual_gate_history]
                all_results['residual_gate_history_b'] = [gate[1] for gate in self.residual_gate_history]
```

```
                all_results['residual_gate_history_a_relative'] = [gate[0]-self.
↪residual_gate_history[0][0] for gate in self.residual_gate_history]
                all_results['residual_gate_history_b_relative'] = [gate[1]-self.
↪residual_gate_history[0][1] for gate in self.residual_gate_history]
                # 标量门控的 _mean 就是自身，便于与向量门控混合绘图
                all_results['residual_gate_history_a_mean'] =␣
↪all_results['residual_gate_history_a']
                all_results['residual_gate_history_b_mean'] =␣
↪all_results['residual_gate_history_b']
                all_results['residual_gate_history_a_mean_relative'] =␣
↪all_results['residual_gate_history_a_relative']
                all_results['residual_gate_history_b_mean_relative'] =␣
↪all_results['residual_gate_history_b_relative']
                all_results['final_residual_gate_a'] = self.
↪residual_gate_values[0]
                all_results['final_residual_gate_b'] = self.
↪residual_gate_values[1]
                all_results['final_residual_gate_a_mean'] =␣
↪all_results['final_residual_gate_a']
                all_results['final_residual_gate_b_mean'] =␣
↪all_results['final_residual_gate_b']
            elif len(self.residual_gate_values.shape) == 2:
                all_results['residual_gate_history_a_mean'] = [gate[:,0].mean()␣
↪for gate in self.residual_gate_history]
                all_results['residual_gate_history_b_mean'] = [gate[:,1].mean()␣
↪for gate in self.residual_gate_history]
                all_results['residual_gate_history_a_mean_relative'] = [gate[:
↪,0].mean()-self.residual_gate_history[0][:,0].mean() for gate in self.
↪residual_gate_history]
                all_results['residual_gate_history_b_mean_relative'] = [gate[:
↪,1].mean()-self.residual_gate_history[0][:,1].mean() for gate in self.
↪residual_gate_history]
                all_results['residual_gate_history_a_std'] = [gate[:,0].std()␣
↪for gate in self.residual_gate_history]
```

```
                all_results['residual_gate_history_b_std'] = [gate[:,1].std()␣
↪for gate in self.residual_gate_history]
                all_results['final_residual_gate_a_mean'] = self.
↪residual_gate_values[:,0].mean()
                all_results['final_residual_gate_b_mean'] = self.
↪residual_gate_values[:,1].mean()
                all_results['final_residual_gate_a_std'] = self.
↪residual_gate_values[:,0].std()
                all_results['final_residual_gate_b_std'] = self.
↪residual_gate_values[:,1].std()
        invalid_keys = [key for key in result_list if key not in all_results]
        if invalid_keys:
            print(f"Warning: The following requested results are not available␣
↪and will be ignored: {invalid_keys}")
        if result_list is not None:
            return {key: all_results[key] for key in result_list if key in␣
↪all_results}

    def offload_to_cpu(self):
        self.model.to('cpu')
        for state in self.optimizer.state.values():
            for k, v in state.items():
                if isinstance(v, torch.Tensor):
                    state[k] = v.cpu()
        self.is_offloaded = True

    def load_to_device(self):
        self.model.to(self.device)
        for state in self.optimizer.state.values():
            for k, v in state.items():
                if isinstance(v, torch.Tensor):
                    state[k] = v.to(self.device)
        self.is_offloaded = False
```

1.2.9 默认配置参数

[119]:
```
def default_setup(manual=None):
    init_parameters = {
    # RoPE/MS-UPE
        'b_rope_or_upe': 10000,             # float: RoPE 或 UPE 的基数
        'head_ratio_upe': 2,                # float: UPE 头比例

    # MultiHeadAttention
        'num_heads': 8,                     # int: 注意力头数 H
        'd_model': 256,                     # int: Transformer 的维度 D
        'max_seq_len': 100,                 # int: 模型支持的最大序列长度（位置编码
相关）
        'pe_type': ['learned_ape'],         # list: 位置编码类型 ('ape',␣
 ↪'learned_ape', 'rope', 'ms_upe', 'alibi')

    # TransformerBlock
        'norm_type': 'layernorm',           # str: transformer_block 中的归一化类
型 ('layernorm' 或 'rmsnorm')
        'ffn_type': 'gelu',                 # str: transformer_block 中前馈网络激
活函数类型 ('gelu' 或 'swiglu')

    # ToyModel
        'num_blocks': 20,                   # int: Transformer 的层数 b
        'loop': True,                       # bool: 是否权重共享
        'residual_gate': (1,1),             # tuple or str: transformer_block 之
间传递残差的门控参数初始值，使`x=a*x+b*x_0` ((a,b) 或'random')
        'residual_gate_type': 'fixed',      # str: 残差门控类型 ('fixed',␣
 ↪'learnable_scalar', 'learnable_vector')
        'residual_random': (1,0.1),          # tuple (mean, std): 当␣
 ↪residual_gate='random' 时满足高斯分布 N(mean, std)

    # RegressionHead
        'd_x': 20,                          # int: 数据输入的特征维度
        'd_y': 1,                           # int: 数据输出的特征维度
        'bias': False,                      # bool: Regression Head 是否使用偏置
```

```
        'init_scale': None,                 # float or None: Regression Head 的初始化缩放因子 (None 表示使用默认初始化)

    # PredictionLoss
        'loss_type': 'mse',                 # str: 损失函数类型 ('mse' 或 'l1')
        'layer_weight_decay': 0.8,          # float: 层权重衰减因子，控制不同层的损失权重 (默认 0.8 表示从最后一层到第一层递减的权重，1.0 表示不加权)
        'seq_weight_decay': 0.8,            # float: 序列权重衰减因子，控制不同时间步的损失权重 (默认 0.8 表示从最后一次预测到第一次预测递减的权重，1.0 表示不加权)

    # LoopedTransformerExperiment
        'lr': 1e-4,                         # float: 学习率
        'gate_lr_ratio': 100,               # float: 当使用残差门控时，门控参数的学习率相对于模型其他参数的倍数
        'seed': 42,                         # int or None: 手动随机种子，确保实验可复现
        'experiment_name': 'Experiment',    # str or None: 实验名称 (None 表示不打印设备和实验初始化信息)
        'timing': True,                     # bool: 是否打印初始化时间
        'optimizer_type': 'adamw',          # str: 优化器类型 ('adam' 或 'sgd'或'adamw')
        'wd_adamw': 0.01,                   # float: AdamW 优化器的权重衰减系数，仅当 optimizer_type='adamw'时有效
        'task': 'regression',               # str: 任务类型 ('regression')
    }

    train_parameters = {
    # MultiHeadAttention
        'batch_size': 64,                   # int: 数据批次大小
        'seq_len': 80,                      # int: 数据序列长度

    # ToyModel
        'num_eff': 15,                      # int: 有效层数，即参与误差计算、参与梯度反向传播的层数 T=b-b_0
```

```
    # dataloader
        'sink_padding': None,               # int or None: seq 最前面填充的 sink token 组数 (None 表示不使用 sink padding)
        'd_hidden': 64,                     # int: 非线性数据生成器中隐藏层的维度，仅当 data_type='nonlinear'时有效
        'function_callable': lambda x: F.relu(x), # function: 非线性数据生成器中使用的非线性函数，仅当 data_type='nonlinear'时有效

    # LoopedTransformerExperiment
        'epochs': 100,                      # int: 训练轮数
        'data_type': 'linear',              # str: 回归数据类型 ('linear')
        'scheduler_type': None,             # str or None: 学习率调节器类型 ('cosine' 或 'step' 或 None)
        'lr_scale': 0.1,                    # float: 学习率调节器的缩放因子，仅当 scheduler_type 不为 None 时有效
        'step_size_scheduler': 10,          # int: StepLR 调度器的步长，仅当 scheduler_type='step'时有效
        'scheduled_training': False,        # bool: 是否使用计划训练，即随着训练的进行逐渐增加参与的层数 b=epoch+num_eff ，直到达到最大层数 num_blocks
        'print_every': 20,                  # int or None: 打印频率/epoch (None 表示不打印训练过程中的损失信息)
        'timing': True,                     # bool: 是否打印训练时间
    }
    if manual is not None:
        for key,value in manual.items():
            if key in init_parameters:
                init_parameters[key]=value
            elif key in train_parameters:
                train_parameters[key]=value
            else:
                print(f"Warning:key {key} do not exist in init_parameters and train_parameters.")
    return init_parameters, train_parameters
```

1.2.10 显存检测

[120]:
```
def print_vram_usage(tag="", peak=None):
    if torch.backends.mps.is_available():
        # MPS 专用 API (Apple Silicon)
        allocated = torch.mps.current_allocated_memory() / (1024 ** 3)
        driver_alloc = torch.mps.driver_allocated_memory() / (1024 ** 3)
        if peak is not None:
            print(f"[{tag}] 显存占用 -> PyTorch 分配: {peak[0]:.2f} GB | Metal
驱动分配: {peak[1]:.2f} GB")
        else:
            print(f"[{tag}] 显存占用 -> PyTorch 分配: {allocated:.2f} GB | Metal
驱动分配: {driver_alloc:.2f} GB")

    elif torch.cuda.is_available():
        # CUDA 专用 API
        allocated = torch.cuda.memory_allocated() / (1024 ** 3)
        peak = torch.cuda.max_memory_allocated() / (1024 ** 3)
        reserved = torch.cuda.memory_reserved() / (1024 ** 3)
        total = torch.cuda.get_device_properties(0).total_memory / (1024 ** 3)
        free = total - reserved
        print(f"[{tag}] 显存占用 -> 已分配: {allocated:.2f} GB | 峰值: {peak:.
↪2f} GB | 缓存池: {reserved:.2f} GB | 剩余可用: {free:.2f} GB")

    else:
        print(f"[{tag}] 当前运行在 CPU，占用主板内存。")
```

1.3 实验台搭建

[ ]:
```
class ExperimentTable:
    '''
    Attributes:
        num_experiments: int, 实验数量
        init_parameters: list of dict, 在 __init__() 方法中传入，用于指定每个实验
的初始化参数，可以覆盖默认参数
        train_parameters: list of dict, 在 __init__() 方法中传入，用于指定每个实
验的训练参数，可以覆盖默认参数
```

```
        result_lists: list of tuple of list, 在 run() 方法中传入，用于指定需要展
示的结果名称和对应的横坐标类型。

        results_groups: list of tuple of list, 包含不同模式
(如'train'和'evaluate') 下的结果列表，每个结果列表是一个包含所有实验结果的列表，
        如 [('train',[experiment1_train_results, experiment2_train_results, ...
 ↪]), ('evaluate', [experiment1_evaluate_results,␣
 ↪experiment2_evaluate_results, ...])]
    '''
    def __init__(self, params_groups:list[dict], manual=None):
        '''
        Args:
            params_groups: list of dict, 每个 dict 包含一个实验的参数设置，可以覆
盖默认参数，
            如 [{residual_gate: (0.5, 0.5), experiment_name: 'Experiment 1'},␣
 ↪{residual_gate: (0.8, 0.2), experiment_name: 'Experiment 2'}]

            manual: dict, 全局默认参数修改

        Attrbutes:
            num_experiments: int, 实验数量
            init_parameters: list of dict, 每个 dict 包含一个实验的初始化参数设置
            train_parameters: list of dict, 每个 dict 包含一个实验的训练参数设置
        '''
        init, train = default_setup(manual=manual)
        self.num_experiments = len(params_groups)
        self.init_parameters = [copy.deepcopy(init) for _ in range(self.
 ↪num_experiments)]
        self.train_parameters = [copy.deepcopy(train) for _ in range(self.
 ↪num_experiments)]
        # 自动分类传入的参数以更新默认参数
        for i, params in enumerate(params_groups):
            for key, value in params.items():
                if key in self.init_parameters[i]:
                    self.init_parameters[i][key] = value
                elif key in self.train_parameters[i]:
```

```
                    self.train_parameters[i][key] = value
                else:
                    raise ValueError(f"Unknown parameter: {key}")

    def run(self, result_lists:list[tuple[list[str], str, int]],␣
↪modes=['train'], parallel_workers=1):
        '''
        Args:
            result_lists: list of tuple of list, 每个 tuple 包含一组实验测量的数
据、对应的横坐标类型和 baseline 索引（不写表示不使用 baseline），每个 list 中的字符
串表示需要展示的结果名称，必须是 LoopedTransformerExperiment.get_results() 中返回
的结果名称，
            如 [(['loss_history', 'residual_gate_history_a_mean',␣
↪'residual_gate_history_b_mean'], 'epoch', 0), (['final_loss',␣
↪'final_residual_gate_a_mean', 'final_residual_gate_b_mean'], 'experiment')]

            modes: list of str, 表示需要运行的模式，可以是 'train' 和 'evaluate'␣
↪的任意组合，'train' 表示需要训练模型并记录训练结果，'evaluate' 表示需要在训练完成
后进行评估并记录评估结果

            parallel_workers: int, 表示并行运行实验的工作线程数。

        Attributes:
            result_lists: 同 Args
            results_groups: dict, 包含不同模式（如'train'和'evaluate'）下的结果列
表，每个 value 是一个 list of dict，每个 dict 是一个实验的结果，为 metric 和 value
的键值对。
            experiments: list of LoopedTransformerExperiment 对象，已转移到 CPU
以节省 GPU 显存
        '''
        # 创建实验对象并运行
        result_set = set().union(*[item[0] for item in result_lists]) # 兼容长度
为 2 或 3 的 tuple，获取所有需要展示的结果名称的集合
        self.result_lists = result_lists
        self.results_groups = {'train': [None]*self.num_experiments, 'evaluate':
↪ [None]*self.num_experiments}
```

```
        self.experiments=[None]*self.num_experiments
        print_lock = threading.Lock()  # 用于在线程中安全地打印日志
        vram_printed = False
        for i in range(self.num_experiments):
            experiment = LoopedTransformerExperiment(**self.init_parameters[i])
            experiment.offload_to_cpu() # 初始化后将模型移回 CPU，节省 GPU 显存
            self.experiments[i] = experiment
        def single_train(i, parallel=False):
            nonlocal vram_printed
            experiment = self.experiments[i]
            experiment.load_to_device() # 训练前将模型加载到 GPU
            if parallel:
                self.train_parameters[i]['timing']=False
                self.train_parameters[i]['print_every']=None
            if parallel:
                with print_lock:
                    if not vram_printed:
                        print_vram_usage(tag=f"Parallel 峰值显存检测")
                        vram_printed = True
            else:
                print_vram_usage(tag=f"Experiment {i+1} 训练前")
            for mode in modes:
                if mode == 'train':
                    experiment.train(**self.train_parameters[i])
                    self.results_groups['train'][i] = experiment.
↪get_results(result_set)
                    if parallel:
                        with print_lock:
                            print_vram_usage(tag=f"Experiment {i+1} 训练峰值",␣
↪peak=(experiment.max_torch_vram, experiment.max_metal_vram))
                    else:
                        print_vram_usage(tag=f"Experiment {i+1} 训练峰值",␣
↪peak=(experiment.max_torch_vram, experiment.max_metal_vram))
                elif mode == 'evaluate':
```

```
                    eval_parameters = {k:v for k,v in self.train_parameters[i].
↪items() if k in ['batch_size', 'seq_len', 'data_type', 'sink_padding',␣
↪'d_hidden', 'function_callable']}
                    if 'loss_type' in self.init_parameters[i]:
                        eval_parameters['loss_type'] = self.
↪init_parameters[i]['loss_type'] # 确保评估使用与训练相同类型的损失函数
                    self.results_groups['evaluate'][i] = experiment.
↪evaluate(**eval_parameters)
            experiment.offload_to_cpu() # 训练完成后将模型和结果移回 CPU，节省 GPU
显存
            self.experiments[i] = experiment
            if torch.cuda.is_available():
                torch.cuda.empty_cache()
            if torch.backends.mps.is_available():
                torch.mps.empty_cache()
            if not parallel:
                print_vram_usage(tag=f"Experiment {i+1} 清理后")
                print(f"Experiment {i+1}/{self.num_experiments} completed.\n")

        print_vram_usage(tag=f"本底显存检测")
        if parallel_workers == 1:
            for i in range(self.num_experiments):
                single_train(i, parallel=False)
        elif isinstance(parallel_workers, int) and parallel_workers > 1:
            print(f"Running {self.num_experiments} experiments in parallel with␣
↪{parallel_workers} workers...")
            with concurrent.futures.
↪ThreadPoolExecutor(max_workers=parallel_workers) as executor:
                executor.map(lambda i: single_train(i, parallel=True),␣
↪range(self.num_experiments))
        else:
            raise ValueError("parallel_workers must be a positive integer.")

    def plot(self, compare_experiments=True,
             subplot_shape=(1,-1),figure_size=None,suptitle='Looped Transformer␣
↪Experiment Results',
```

```
            colors=['blue', 'red', 'green', 'yellow', 'cyan', 'orange',␣
↪'purple', 'brown']):
        '''
        Args:
            compare_experiments: bool, True 表示在一张图中比较不同实验的结果，每张图仅比较一个指标；False 表示每个实验单独成图，每张图仅比较一个实验的多个指标（第一个指标使用左轴，第二个及以后的指标使用共享 x 轴的右轴）

            subplot_shape: tuple of int, 表示子图的行数和列数，(-1 表示自动计算)，如 (1, -1) 表示所有图在一行，(-1, 2) 表示每行两个图，自动计算行数

            figure_size: tuple of float, 画布大小，单位为英寸，如 (15, 10)。默认值 None 表示使用 (8,6)*subplot_shape 的大小

            suptitle: str, 整体图的标题

            colors: list of str, 画图使用的颜色列表，每张图从头开始循环使用这些颜色
        '''
        # 画图
        if compare_experiments:
            num_plots = len(self.result_lists) # 需要画的图的总数量
        else:
            num_plots = len(self.result_lists)*self.num_experiments # 需要画的图的总数量
        if subplot_shape == (-1, -1):
            raise ValueError("Invalid subplot_shape: both dimensions cannot be␣
↪-1. Please specify at least one dimension or provide a valid shape.")
        if subplot_shape[1] == -1:
            subplot_shape = (subplot_shape[0], math.ceil(num_plots /␣
↪subplot_shape[0]))  # 自动计算列数
        elif subplot_shape[0] == -1:
            subplot_shape = (math.ceil(num_plots / subplot_shape[1]),␣
↪subplot_shape[1])  # 自动计算行数
        if subplot_shape[0] * subplot_shape[1] < num_plots:
            raise ValueError(f"subplot_shape {subplot_shape} is too small for␣
↪the number of plots {num_plots}. Please increase the dimensions.")
```

```
        if figure_size is None:
            figure_size = (8*subplot_shape[1], 6*subplot_shape[0])

        if self.results_groups['train'][0] is not None:
            results = self.results_groups['train']
            fig,axs = plt.subplots(*subplot_shape, figsize=figure_size)
            axs_flat = np.atleast_1d(axs).flatten()  # 将 axs 转换为一维数组，方便索引
            index=0
            for item in self.result_lists:
                result_list = item[0]  # 需要展示的结果名称列表
                x = item[1]            # 横坐标类型 ('epoch' 或 'experiment')
                baseline_index = item[2] if len(item) > 2 else None  # 基线索引(可选)，表示用于计算相对值的基线实验索引
                baseline_name=None
                if x=='epoch': # 表示画横坐标是 epoch 的曲线
                    if compare_experiments:
                        for metric in result_list:
                            if baseline_index is not None and 0 <=␣
 ↪baseline_index < self.num_experiments:
                                baseline_name = self.
 ↪init_parameters[baseline_index].get('experiment_name', f'Experiment␣
 ↪{baseline_index+1}')
                                baseline_data = np.
 ↪array(results[baseline_index][metric])
                            for i in range(self.num_experiments):
                                exp_name = self.init_parameters[i].
 ↪get('experiment_name', f'Experiment {i+1}')
                                y_data = np.array(results[i][metric])
                                if baseline_name is not None:
                                    y_data = y_data - baseline_data  # 计算相对于基线的差值
                                    if i == baseline_index:
                                        exp_name += ' (Baseline)'
                                axs_flat[index].plot(y_data, label=exp_name,␣
 ↪color=colors[i%len(colors)])
```

```
                            axs_flat[index].set_xlabel('Epoch')
                            if baseline_name is not None:
                                axs_flat[index].set_ylabel(f"{metric} (relative␣
↪to {baseline_name})")
                            else:
                                axs_flat[index].set_ylabel(metric)
                            axs_flat[index].tick_params(axis='y')
                            axs_flat[index].set_title('Comparison of ' + metric)
                            axs_flat[index].legend(loc='upper right')
                            axs_flat[index].grid(True)
                            index += 1
                    else:
                        if baseline_index is not None and 0 <= baseline_index <␣
↪self.num_experiments:
                            baseline_name = self.
↪init_parameters[baseline_index].get('experiment_name', f'Experiment␣
↪{baseline_index+1}')
                            baseline_data = np.
↪array(results[baseline_index][result_list[0]])
                        for i in range(self.num_experiments):
                            color_index = 0
                            y_data = np.array(results[i][result_list[0]])
                            if baseline_name is not None:
                                y_data = y_data - baseline_data  # 计算相对于基
线的差值
                                if i == baseline_index:
                                    exp_name += ' (Baseline)'
                            axs_flat[index].plot(y_data, label=result_list[0],␣
↪color=colors[color_index%len(colors)])
                            axs_flat[index].set_xlabel('Epoch')
                            if baseline_name is not None:
                                axs_flat[index].set_ylabel(f"{result_list[0]}␣
↪(relative to {baseline_name})", color=colors[color_index%len(colors)])
                            else:
                                axs_flat[index].set_ylabel(result_list[0],␣
↪color=colors[color_index%len(colors)])
```

```
                            axs_flat[index].tick_params(axis='y',␣
↪labelcolor=colors[color_index%len(colors)])
                            color_index += 1
                            if len(result_list) > 1:
                                ax_twin = axs_flat[index].twinx()  # 创建共享 x
轴但独立 y 轴的第二个坐标轴
                                primary_color = colors[color_index%len(colors)]
                                for j in range(1, len(result_list)):
                                    y_data = np.
↪array(results[i][result_list[j]])
                                    if baseline_name is not None:
                                        y_data = y_data - baseline_data  # 计算
相对于基线的差值
                                        if i == baseline_index:
                                            exp_name += ' (Baseline)'
                                    ax_twin.plot(y_data, label=result_list[j],␣
↪color=colors[color_index%len(colors)])
                                    color_index += 1
                                ax_twin.set_ylabel(' and '.join(result_list[1:
↪]), color=primary_color)
                                ax_twin.tick_params(axis='y',␣
↪labelcolor=primary_color)
                                ax_twin.legend(loc='upper right')
                            axs_flat[index].set_title(self.
↪init_parameters[i]['experiment_name']+'-'+' and '.join(result_list))
                            axs_flat[index].legend(loc='upper left')
                            axs_flat[index].grid(True)
                            index += 1
                elif x == 'experiment': # 表示画不同实验对比的柱状图
                    exp_names = [self.init_parameters[i].get('experiment_name',␣
↪f'Experiment {i+1}') for i in range(self.num_experiments)]
                    if baseline_index is not None and 0 <= baseline_index <␣
↪self.num_experiments:
                        baseline_name = self.init_parameters[baseline_index].
↪get('experiment_name', f'Experiment {baseline_index+1}')
```

```
                    exp_names[baseline_index] += '(Baseline)'

                x_pos = np.arange(len(exp_names))
                num_metrics = len(result_list)
                width = 0.8 / num_metrics # 确保所有柱子加起来不超过 0.8 的宽度，留下 0.2 的组间空白
                color_index = 0
                metric0 = result_list[0]
                y_data0 = [results[i].get(metric0, 0) for i in range(self.num_experiments)]
                if baseline_name is not None:
                    base_val = y_data0[baseline_index]
                    y_data0 = [val - base_val for val in y_data0]

                offset0 = (0 - num_metrics / 2 + 0.5) * width
                axs_flat[index].bar(x_pos + offset0, y_data0, width, label=metric0, color=colors[color_index % len(colors)])
                ylabel0 = f"{metric0}\n(relative to {baseline_name})" if baseline_name else metric0
                axs_flat[index].set_ylabel(ylabel0, color=colors[color_index % len(colors)])
                axs_flat[index].tick_params(axis='y', labelcolor=colors[color_index % len(colors)])
                color_index += 1
                if num_metrics > 1:
                    ax_twin = axs_flat[index].twinx() # 创建共享 X 轴但独立 Y 轴的坐标系

                    primary_color = colors[color_index % len(colors)]
                    for j in range(1, num_metrics):
                        metric_j = result_list[j]
                        y_data_j = [results[i].get(metric_j, 0) for i in range(self.num_experiments)]
                        if baseline_name is not None:
                            base_val = y_data_j[baseline_index]
                            y_data_j = [val - base_val for val in y_data_j]
```

```
                        offset_j = (j - num_metrics / 2 + 0.5) * width
                        ax_twin.bar(x_pos + offset_j, y_data_j, width,␣
↪label=metric_j, color=colors[color_index % len(colors)])
                        color_index += 1
                    ylabel_right = ' and '.join(result_list[1:])
                    if baseline_name is not None:
                        ylabel_right += f"\n(relative to {baseline_name})"

                    ax_twin.set_ylabel(ylabel_right, color=primary_color)
                    ax_twin.tick_params(axis='y', labelcolor=primary_color)
                    ax_twin.legend(loc='upper right')
                    axs_flat[index].legend(loc='upper left')
                else:
                    axs_flat[index].legend(loc='best')
                y1_min, y1_max = axs_flat[index].get_ylim()
                pos1 = max(y1_max, 0)
                neg1 = max(-y1_min, 0)
                s1 = max(pos1, neg1) if max(pos1, neg1) > 0 else 1.0
                pos_norm1 = pos1 / s1
                neg_norm1 = neg1 / s1
                if num_metrics > 1:
                    y2_min, y2_max = ax_twin.get_ylim()
                    pos2 = max(y2_max, 0)
                    neg2 = max(-y2_min, 0)
                    s2 = max(pos2, neg2) if max(pos2, neg2) > 0 else 1.0
                    pos_norm2 = pos2 / s2
                    neg_norm2 = neg2 / s2
                    max_pos_norm = max(pos_norm1, pos_norm2)
                    max_neg_norm = max(neg_norm1, neg_norm2)
                else:
                    max_pos_norm = pos_norm1
                    max_neg_norm = neg_norm1
                    s2 = 1.0 # 占位
                final_pos_norm = max(max_pos_norm * 1.35, 0.35)
                needs_zero_alignment = (baseline_name is not None) or␣
↪(y1_min < 0) or (num_metrics > 1 and y2_min < 0)
```

```
                    if needs_zero_alignment:
                        # 按照统一计算好的归一化比例，乘以各自的真实尺度
                        axs_flat[index].set_ylim(-max_neg_norm * s1,␣
↪final_pos_norm * s1)
                        if num_metrics > 1:
                            ax_twin.set_ylim(-max_neg_norm * s2, final_pos_norm␣
↪* s2)
                    else:
                        axs_flat[index].set_ylim(0, final_pos_norm * s1)
                        if num_metrics > 1:
                            ax_twin.set_ylim(0, final_pos_norm * s2)
                    axs_flat[index].axhline(0, color='gray', linewidth=1,␣
↪linestyle='-', zorder=0) # 画一条贯穿整个图表的辅助零线，充当视觉上的“地平线”
                    axs_flat[index].set_xticks(x_pos)
                    axs_flat[index].set_xticklabels(exp_names, rotation=45,␣
↪ha='right')
                    axs_flat[index].set_title(f"Comparison of {' and '.
↪join(result_list)}")
                    axs_flat[index].grid(axis='y', linestyle='--', alpha=0.7)
                    index += 1
            # 如果绘制的图的数量少于子图的总数量，则隐藏多余的子图
            for i in range(index, len(axs_flat)):
                axs_flat[i].set_visible(False)
        if self.results_groups['evaluate'][0] is not None:
            pass
        if suptitle is not None:
            plt.suptitle(suptitle)
        plt.tight_layout()
        plt.show()
```

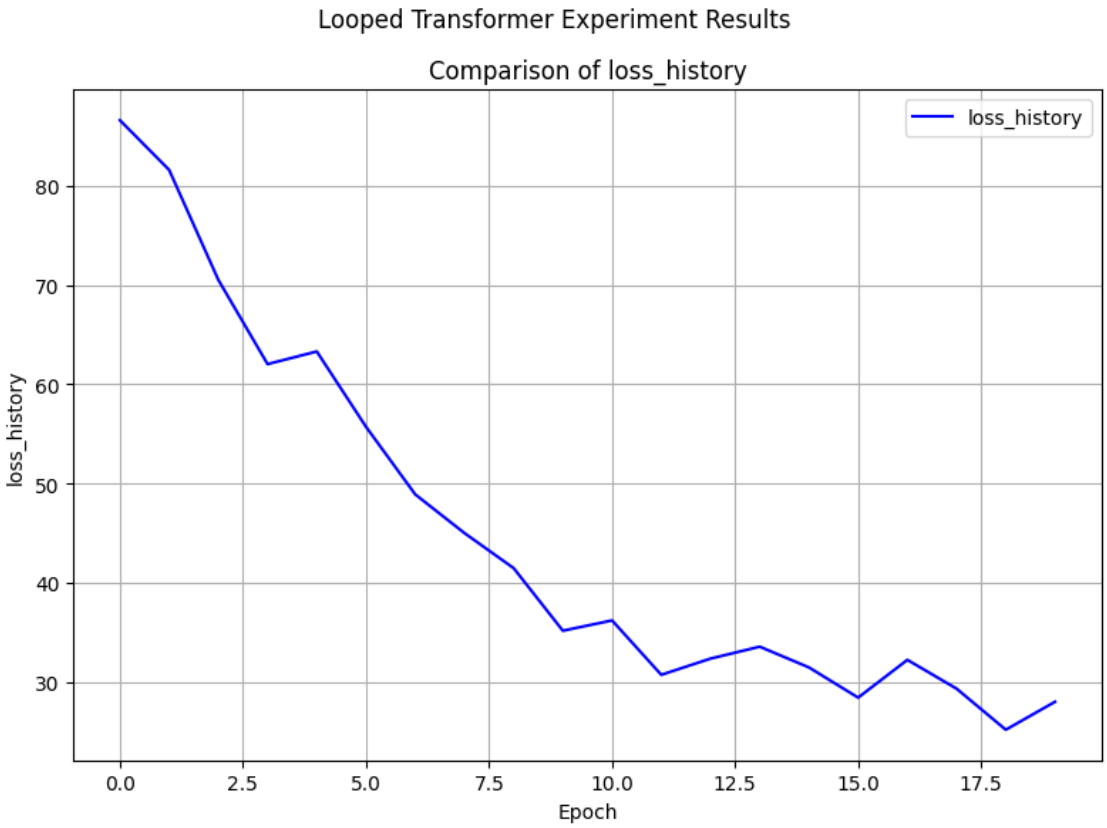
2 Linear Regression Experiments

2.1 实验数据观测

2.1.1 绘制 Loss 下降曲线

[23]:
```
loss_table = ExperimentTable(params_groups=[{'experiment_name':␣
↪'loss_history'}])
loss_table.run(result_lists=[(['loss_history'], 'epoch')])
loss_table.plot(figure_size=(8,6))
```

```
loss_history initialized in 0.02 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.02 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.02 GB
Epoch 5/20, Loss: 63.3171, current_blocks: 19
Epoch 10/20, Loss: 35.1716, current_blocks: 20
Epoch 15/20, Loss: 31.4720, current_blocks: 20
Epoch 20/20, Loss: 28.0077, current_blocks: 20
Training completed in 6.75 seconds.
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.15 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 1/1 completed.
```

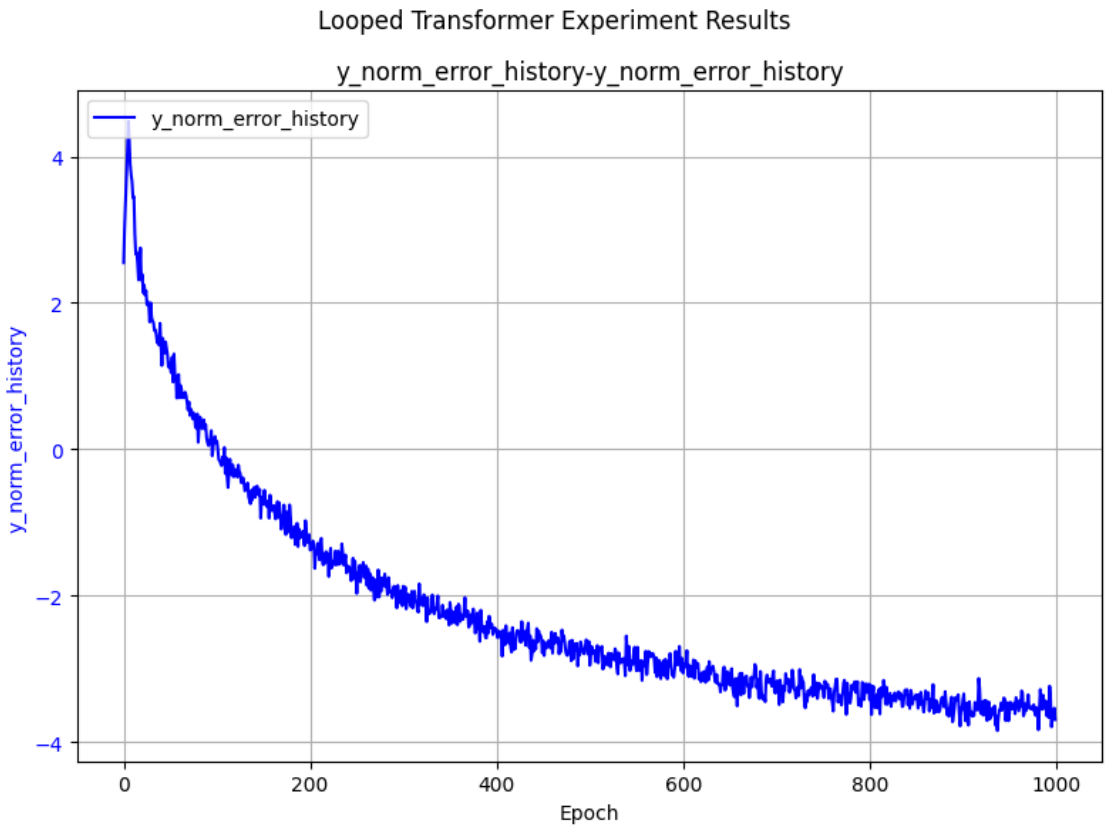


2.1.2 绘制 y_pred 和 y_true 的变化曲线

[75]:
```
norm_table = ExperimentTable(params_groups=[{'epochs': 1000, 'experiment_name':␣
↪'y_norm_error_history', 'print_every': 200}])
norm_table.run(result_lists=[(['y_norm_error_history'], 'epoch')])
norm_table.plot(figure_size=(8,6),compare_experiments=False)
```

```
y_norm_error_history initialized in 0.02 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Epoch 200/1000, Loss: 18.8248, current_blocks: 20
Epoch 400/1000, Loss: 19.7988, current_blocks: 20
Epoch 600/1000, Loss: 17.7491, current_blocks: 20
Epoch 800/1000, Loss: 20.9858, current_blocks: 20
Epoch 1000/1000, Loss: 20.9834, current_blocks: 20
Training completed in 451.35 seconds.
```

```
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Experiment 1/1 completed.
```



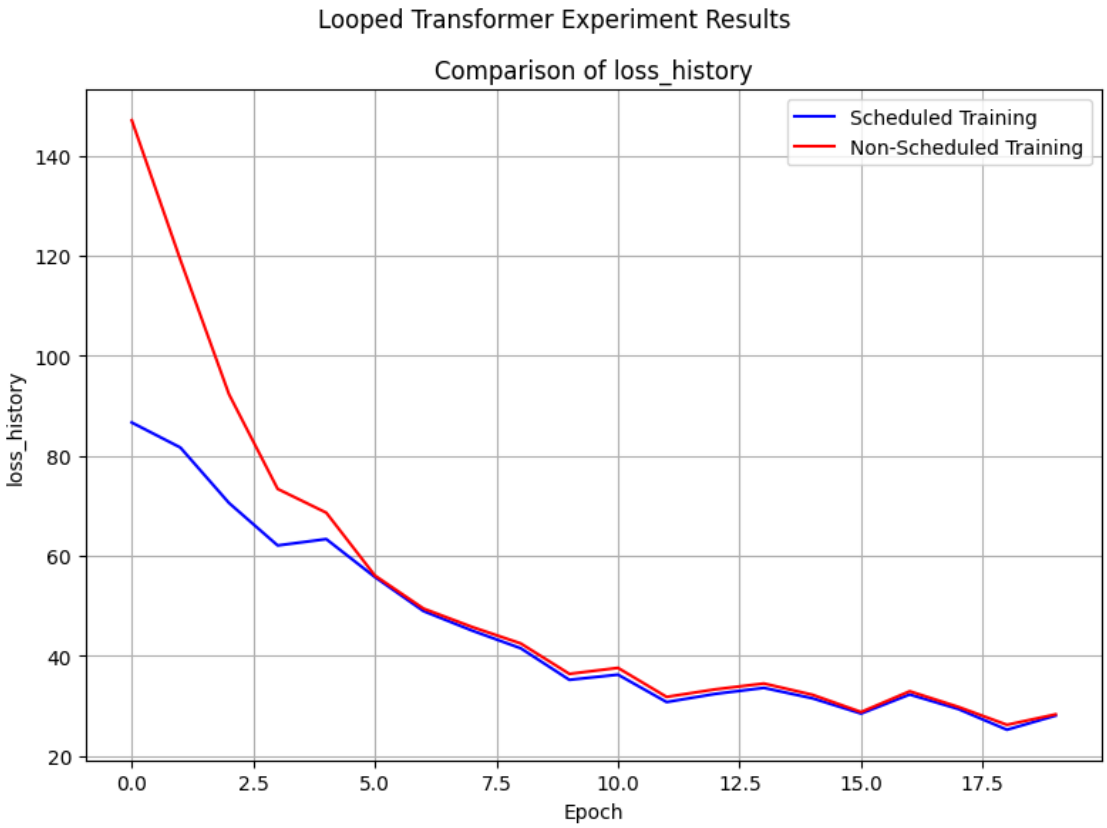
2.2 对比实验

2.2.1 Scheduled Training vs Non-Scheduled Training 对比实验

[76]:
```
table = ExperimentTable(params_groups=[{'scheduled_training': True,␣
↪'experiment_name': 'Scheduled Training'},
                                       {'scheduled_training': False,␣
↪'experiment_name': 'Non-Scheduled Training'}])
table.run(result_lists=[(['loss_history'], 'epoch')])
table.plot(figure_size=(8, 6), compare_experiments=True)
```

```
Scheduled Training initialized in 0.02 seconds. Using device: mps
```

```
Non-Scheduled Training initialized in 0.01 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.05 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.05 GB
Epoch 5/20, Loss: 63.3171, current_blocks: 19
Epoch 10/20, Loss: 35.1716, current_blocks: 20
Epoch 15/20, Loss: 31.4720, current_blocks: 20
Epoch 20/20, Loss: 28.0077, current_blocks: 20
Training completed in 9.75 seconds.
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Experiment 1/2 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.05 GB
Epoch 5/20, Loss: 68.5955, current_blocks: 20
Epoch 10/20, Loss: 36.3633, current_blocks: 20
Epoch 15/20, Loss: 32.1849, current_blocks: 20
Epoch 20/20, Loss: 28.2660, current_blocks: 20
Training completed in 9.81 seconds.
[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.13 GB
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Experiment 2/2 completed.
```



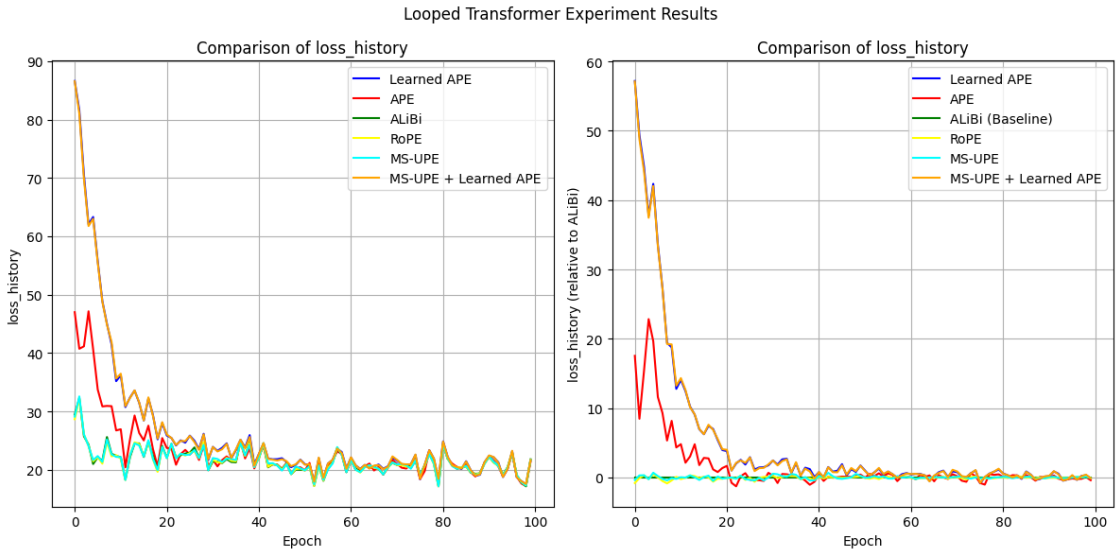
2.2.2 不同 Positional Embedding 对比实验

[79]:
```
pe_table = ExperimentTable(params_groups=[
    {'pe_type': ['learned_ape'], 'experiment_name': 'Learned APE'},
    {'pe_type': ['ape'], 'experiment_name': 'APE'},
    {'pe_type': ['alibi'], 'experiment_name': 'ALiBi'},
    {'pe_type': ['rope'], 'experiment_name': 'RoPE'},
    {'pe_type': ['ms_upe'], 'experiment_name': 'MS-UPE'},
    {'pe_type': ['ms_upe', 'learned_ape'], 'experiment_name': 'MS-UPE + Learned␣
 ↪APE'},
], manual={'epochs': 100, 'print_every': 20})
pe_table.run(result_lists=[
    (['loss_history'], 'epoch'),
    (['loss_history'], 'epoch', 2)  # 以 ALiBi 为基线
])
```

```
pe_table.plot(figure_size=(12, 6))
```

```
Learned APE initialized in 0.02 seconds. Using device: mps
APE initialized in 0.01 seconds. Using device: mps
ALiBi initialized in 0.01 seconds. Using device: mps
RoPE initialized in 0.01 seconds. Using device: mps
MS-UPE initialized in 0.01 seconds. Using device: mps
MS-UPE + Learned APE initialized in 0.01 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.19 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.19 GB
Epoch 20/100, Loss: 28.0077, current_blocks: 20
Epoch 40/100, Loss: 20.8999, current_blocks: 20
Epoch 60/100, Loss: 20.2006, current_blocks: 20
Epoch 80/100, Loss: 18.3025, current_blocks: 20
Epoch 100/100, Loss: 21.6682, current_blocks: 20
Training completed in 47.76 seconds.
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.22 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
Experiment 1/6 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 25.4152, current_blocks: 20
Epoch 40/100, Loss: 20.2102, current_blocks: 20
Epoch 60/100, Loss: 19.7146, current_blocks: 20
Epoch 80/100, Loss: 17.6053, current_blocks: 20
Epoch 100/100, Loss: 21.3824, current_blocks: 20
Training completed in 48.05 seconds.
[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.21 GB
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
Experiment 2/6 completed.

[Experiment 3 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 24.0522, current_blocks: 20
Epoch 40/100, Loss: 20.8352, current_blocks: 20
Epoch 60/100, Loss: 19.5621, current_blocks: 20
Epoch 80/100, Loss: 17.1478, current_blocks: 20
Epoch 100/100, Loss: 21.7769, current_blocks: 20
```

```
Training completed in 55.24 seconds.
[Experiment 3 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.23 GB
[Experiment 3 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
Experiment 3/6 completed.

[Experiment 4 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 23.8768, current_blocks: 20
Epoch 40/100, Loss: 20.4678, current_blocks: 20
Epoch 60/100, Loss: 19.6902, current_blocks: 20
Epoch 80/100, Loss: 17.1698, current_blocks: 20
Epoch 100/100, Loss: 21.6961, current_blocks: 20
Training completed in 68.12 seconds.
[Experiment 4 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.22 GB
[Experiment 4 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
Experiment 4/6 completed.

[Experiment 5 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 23.8994, current_blocks: 20
Epoch 40/100, Loss: 20.4356, current_blocks: 20
Epoch 60/100, Loss: 19.6087, current_blocks: 20
Epoch 80/100, Loss: 17.1837, current_blocks: 20
Epoch 100/100, Loss: 21.6859, current_blocks: 20
Training completed in 52.83 seconds.
[Experiment 5 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.22 GB
[Experiment 5 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
Experiment 5/6 completed.

[Experiment 6 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 28.1398, current_blocks: 20
Epoch 40/100, Loss: 20.6823, current_blocks: 20
Epoch 60/100, Loss: 20.1465, current_blocks: 20
Epoch 80/100, Loss: 18.4156, current_blocks: 20
Epoch 100/100, Loss: 21.6613, current_blocks: 20
Training completed in 54.00 seconds.
[Experiment 6 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.23 GB
[Experiment 6 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
Experiment 6/6 completed.
```



2.2.3 sink padding 的有无对比实验

[80]:
```
sink_table = ExperimentTable(params_groups=[
    {'experiment_name': 'Without Sink Padding'},
    {'experiment_name': 'With Sink Padding (1)', 'sink_padding': 1},
    {'experiment_name': 'With Sink Padding (2)', 'sink_padding': 2},
    {'experiment_name': 'With Sink Padding (5)', 'sink_padding': 5},
    {'experiment_name': 'With Sink Padding (10)', 'sink_padding': 10},
], manual={'epochs': 100, 'print_every': 20})
sink_table.run(result_lists=[
    (['loss_history'], 'epoch'),
    (['loss_history'], 'epoch', 0),  # 以 Without Sink Padding 为基线
])
sink_table.plot(figure_size=(12, 6))
```

```
Without Sink Padding initialized in 0.02 seconds. Using device: mps
With Sink Padding (1) initialized in 0.01 seconds. Using device: mps
With Sink Padding (2) initialized in 0.01 seconds. Using device: mps
With Sink Padding (5) initialized in 0.01 seconds. Using device: mps
With Sink Padding (10) initialized in 0.01 seconds. Using device: mps
```

[本底显存检测] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 28.0077, current_blocks: 20
Epoch 40/100, Loss: 20.8999, current_blocks: 20
Epoch 60/100, Loss: 20.2006, current_blocks: 20
Epoch 80/100, Loss: 18.3025, current_blocks: 20
Epoch 100/100, Loss: 21.6682, current_blocks: 20
Training completed in 50.96 seconds.
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.22 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
Experiment 1/5 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 28.1106, current_blocks: 20
Epoch 40/100, Loss: 20.9680, current_blocks: 20
Epoch 60/100, Loss: 20.6434, current_blocks: 20
Epoch 80/100, Loss: 17.8341, current_blocks: 20
Epoch 100/100, Loss: 21.5738, current_blocks: 20
Training completed in 53.92 seconds.
[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.22 GB
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
Experiment 2/5 completed.

[Experiment 3 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 29.5266, current_blocks: 20
Epoch 40/100, Loss: 20.6409, current_blocks: 20
Epoch 60/100, Loss: 20.1771, current_blocks: 20
Epoch 80/100, Loss: 18.3634, current_blocks: 20
Epoch 100/100, Loss: 21.4438, current_blocks: 20
Training completed in 53.87 seconds.
[Experiment 3 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 5.22 GB
[Experiment 3 清理后] 显存占用 -> PyTorch 分配: 3.10 GB | Metal 驱动分配: 4.21 GB
Experiment 3/5 completed.

[Experiment 4 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 26.8264, current_blocks: 20
Epoch 40/100, Loss: 20.8156, current_blocks: 20

```
Epoch 60/100, Loss: 20.2012, current_blocks: 20
Epoch 80/100, Loss: 18.7726, current_blocks: 20
Epoch 100/100, Loss: 21.6547, current_blocks: 20
Training completed in 57.91 seconds.
[Experiment 4 训练峰值] 显存占用 -> PyTorch 分配: 3.13 GB | Metal 驱动分配: 6.22 GB
[Experiment 4 清理后] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Experiment 4/5 completed.

[Experiment 5 训练前] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Epoch 20/100, Loss: 27.0638, current_blocks: 20
Epoch 40/100, Loss: 21.4043, current_blocks: 20
Epoch 60/100, Loss: 20.5190, current_blocks: 20
Epoch 80/100, Loss: 18.5852, current_blocks: 20
Epoch 100/100, Loss: 21.4375, current_blocks: 20
Training completed in 66.44 seconds.
[Experiment 5 训练峰值] 显存占用 -> PyTorch 分配: 3.14 GB | Metal 驱动分配: 6.26 GB
[Experiment 5 清理后] 显存占用 -> PyTorch 分配: 3.11 GB | Metal 驱动分配: 4.21 GB
Experiment 5/5 completed.
```

2.2.4 Residual Gate 对比实验

[27]:
```
gate_table = ExperimentTable(params_groups=[
    {'experiment_name': 'No Residual Gate', 'residual_gate': (0, 0),␣
 ↪'residual_gate_type': 'fixed'},
    {'experiment_name': 'Residual Gate (1, 0.5)', 'residual_gate': (1, 0.5),␣
 ↪'residual_gate_type': 'learnable_scalar'},
    {'experiment_name': 'Residual Gate (1, 0.5) Vector', 'residual_gate': (1, 0.
 ↪5), 'residual_gate_type': 'learnable_vector'},
    {'experiment_name': 'Residual Gate (1, 0)', 'residual_gate': (1, 0),␣
 ↪'residual_gate_type': 'learnable_scalar'},
    {'experiment_name': 'Residual Gate (1, 0) Vector', 'residual_gate': (1, 0),␣
 ↪'residual_gate_type': 'learnable_vector'},
    {'experiment_name': 'Random Learnable Scalar', 'residual_gate': 'random',␣
 ↪'residual_gate_type': 'learnable_scalar', 'residual_random': (0.9, 0.1)},
    {'experiment_name': 'Random Learnable Vector', 'residual_gate': 'random',␣
 ↪'residual_gate_type': 'learnable_vector', 'residual_random': (0.9, 0.1)},
], manual={'gate_lr_ratio': 100, 'scheduler_type': 'cosine', 'epochs': 50,␣
 ↪'print_every': 25})
gate_table.run(result_lists=[
    (['loss_history'], 'epoch')
])
gate_table.plot(figure_size=(8, 6))
```

```
No Residual Gate initialized in 0.01 seconds. Using device: mps
Residual Gate (1, 0.5) initialized in 0.01 seconds. Using device: mps
Residual Gate (1, 0.5) Vector initialized in 0.01 seconds. Using device: mps
Residual Gate (1, 0) initialized in 0.01 seconds. Using device: mps
Residual Gate (1, 0) Vector initialized in 0.01 seconds. Using device: mps
Random Learnable Scalar initialized in 0.01 seconds. Using device: mps
Random Learnable Vector initialized in 0.01 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/50, Loss: 20.7067, current_blocks: 20
Epoch 50/50, Loss: 19.7848, current_blocks: 20
Training completed in 19.79 seconds.
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
```

[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 1/7 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Epoch 25/50, Loss: 21.0445, current_blocks: 20
Epoch 50/50, Loss: 19.9295, current_blocks: 20
Training completed in 20.28 seconds.
[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.15 GB
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 2/7 completed.

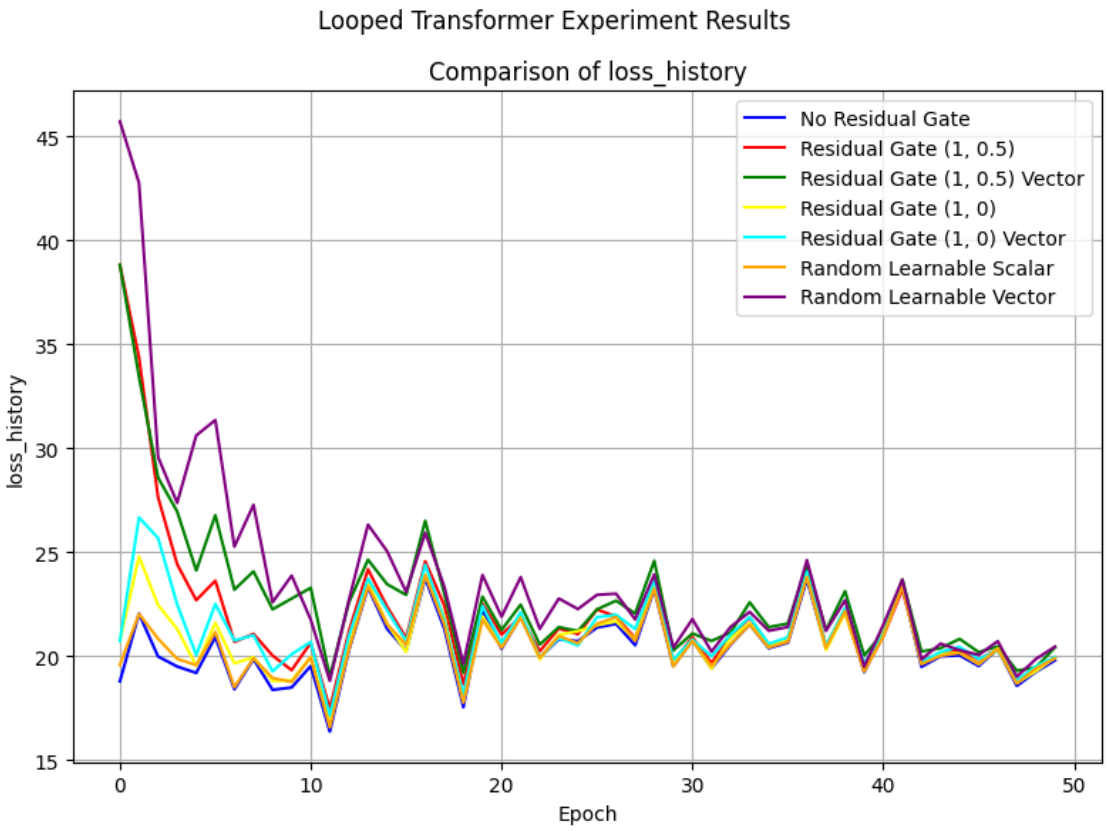
[Experiment 3 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Epoch 25/50, Loss: 21.1970, current_blocks: 20
Epoch 50/50, Loss: 20.4095, current_blocks: 20
Training completed in 20.98 seconds.
[Experiment 3 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.15 GB
[Experiment 3 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 3/7 completed.

[Experiment 4 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/50, Loss: 21.1870, current_blocks: 20
Epoch 50/50, Loss: 19.9217, current_blocks: 20
Training completed in 20.64 seconds.
[Experiment 4 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.15 GB
[Experiment 4 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 4/7 completed.

[Experiment 5 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/50, Loss: 20.5020, current_blocks: 20
Epoch 50/50, Loss: 19.9380, current_blocks: 20
Training completed in 21.49 seconds.
[Experiment 5 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
[Experiment 5 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 5/7 completed.

[Experiment 6 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/50, Loss: 20.6064, current_blocks: 20

```
Epoch 50/50, Loss: 19.8823, current_blocks: 20
Training completed in 21.05 seconds.
[Experiment 6 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
[Experiment 6 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 6/7 completed.

[Experiment 7 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/50, Loss: 22.2538, current_blocks: 20
Epoch 50/50, Loss: 20.4417, current_blocks: 20
Training completed in 21.80 seconds.
[Experiment 7 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
[Experiment 7 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 7/7 completed.
```



[26]:
```
gate_table = ExperimentTable(params_groups=[
    {'experiment_name': 'Residual Gate (1, 0.5)', 'residual_gate': (1, 0.5),␣
 ↪'residual_gate_type': 'learnable_scalar'},
    {'experiment_name': 'Residual Gate (1, 0.5) Vector', 'residual_gate': (1, 0.
 ↪5), 'residual_gate_type': 'learnable_vector'},
    {'experiment_name': 'Residual Gate (1, 0)', 'residual_gate': (1, 0),␣
 ↪'residual_gate_type': 'learnable_scalar'},
    {'experiment_name': 'Residual Gate (1, 0) Vector', 'residual_gate': (1, 0),␣
 ↪'residual_gate_type': 'learnable_vector'},
    {'experiment_name': 'Random Learnable Scalar', 'residual_gate': 'random',␣
 ↪'residual_gate_type': 'learnable_scalar', 'residual_random': (0.9, 0.1)},
    {'experiment_name': 'Random Learnable Vector', 'residual_gate': 'random',␣
 ↪'residual_gate_type': 'learnable_vector', 'residual_random': (0.9, 0.1)},
], manual={'gate_lr_ratio': 100, 'scheduler_type': 'cosine', 'epochs': 100,␣
 ↪'print_every': 25})
gate_table.run(result_lists=[
    (['residual_gate_history_a_mean_relative',␣
 ↪'residual_gate_history_b_mean_relative'], 'epoch')
])
gate_table.plot(figure_size=(20,30),compare_experiments=False,␣
 ↪subplot_shape=(-1,2), suptitle='')
```

```
Residual Gate (1, 0.5) initialized in 0.02 seconds. Using device: mps
Residual Gate (1, 0.5) Vector initialized in 0.01 seconds. Using device: mps
Residual Gate (1, 0) initialized in 0.01 seconds. Using device: mps
Residual Gate (1, 0) Vector initialized in 0.01 seconds. Using device: mps
Random Learnable Scalar initialized in 0.01 seconds. Using device: mps
Random Learnable Vector initialized in 0.01 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.02 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.02 GB
Epoch 25/100, Loss: 21.1192, current_blocks: 20
Epoch 50/100, Loss: 19.8862, current_blocks: 20
Epoch 75/100, Loss: 21.3765, current_blocks: 20
Epoch 100/100, Loss: 20.4947, current_blocks: 20
Training completed in 34.69 seconds.
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
```

[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 1/6 completed.

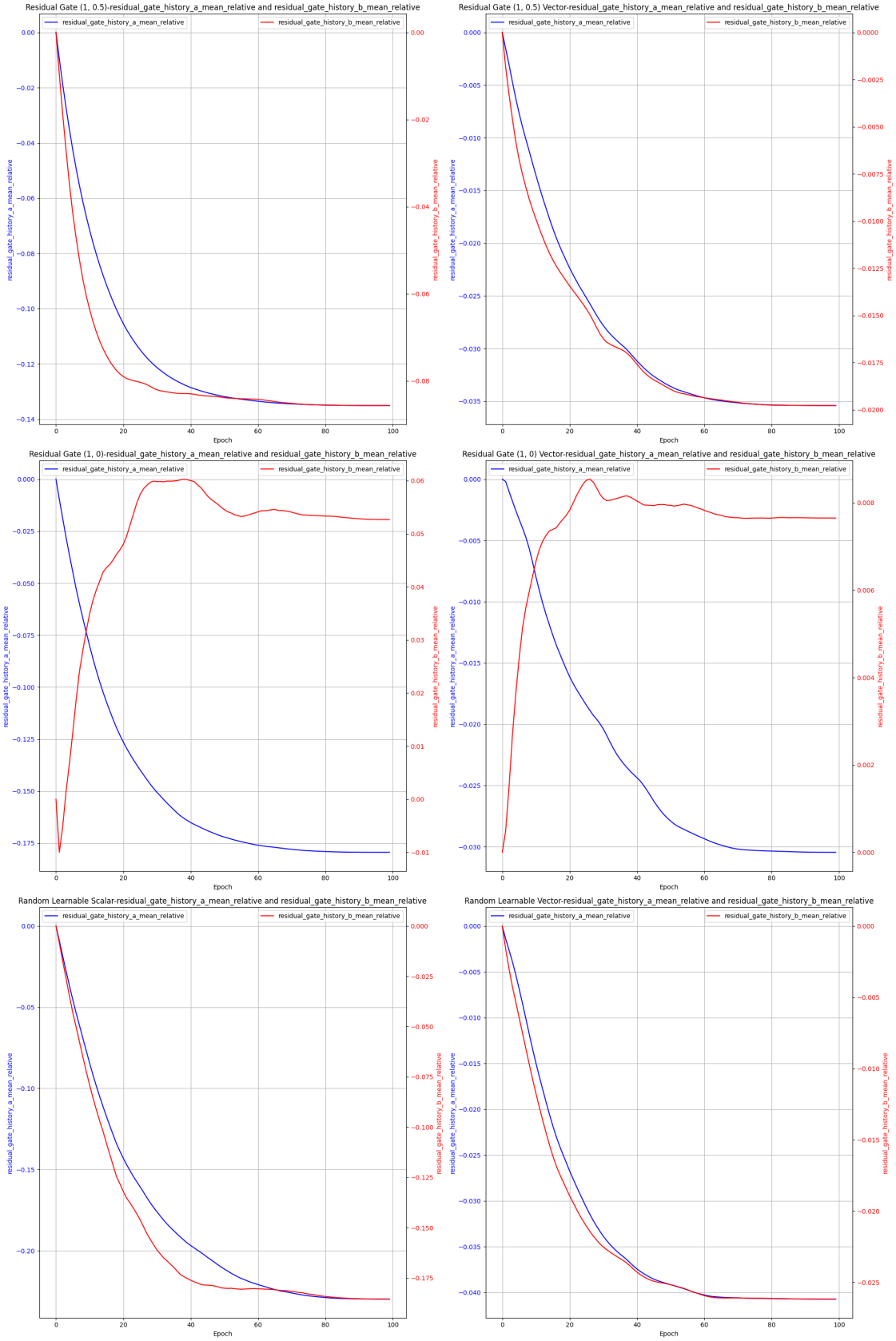
[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/100, Loss: 21.3562, current_blocks: 20
Epoch 50/100, Loss: 20.4467, current_blocks: 20
Epoch 75/100, Loss: 21.5648, current_blocks: 20
Epoch 100/100, Loss: 20.8265, current_blocks: 20
Training completed in 35.89 seconds.
[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.15 GB
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 2/6 completed.

[Experiment 3 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/100, Loss: 21.0529, current_blocks: 20
Epoch 50/100, Loss: 19.8676, current_blocks: 20
Epoch 75/100, Loss: 21.1713, current_blocks: 20
Epoch 100/100, Loss: 20.4512, current_blocks: 20
Training completed in 35.34 seconds.
[Experiment 3 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.15 GB
[Experiment 3 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 3/6 completed.

[Experiment 4 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/100, Loss: 20.5109, current_blocks: 20
Epoch 50/100, Loss: 20.1726, current_blocks: 20
Epoch 75/100, Loss: 21.4211, current_blocks: 20
Epoch 100/100, Loss: 20.5051, current_blocks: 20
Training completed in 36.48 seconds.
[Experiment 4 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.15 GB
[Experiment 4 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 4/6 completed.

[Experiment 5 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/100, Loss: 20.5986, current_blocks: 20
Epoch 50/100, Loss: 19.8349, current_blocks: 20
Epoch 75/100, Loss: 21.2072, current_blocks: 20

```
Epoch 100/100, Loss: 20.3681, current_blocks: 20
Training completed in 37.95 seconds.
[Experiment 5 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
[Experiment 5 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 5/6 completed.

[Experiment 6 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 25/100, Loss: 22.4328, current_blocks: 20
Epoch 50/100, Loss: 20.2457, current_blocks: 20
Epoch 75/100, Loss: 21.3175, current_blocks: 20
Epoch 100/100, Loss: 20.7065, current_blocks: 20
Training completed in 41.02 seconds.
[Experiment 6 训练峰值] 显存占用 -> PyTorch 分配: 0.03 GB | Metal 驱动分配: 2.16 GB
[Experiment 6 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.03 GB
Experiment 6/6 completed.
```


Residual Gate (1, 0.5)-residual_gate_history_a_mean_relative and residual_gate_history_b_mean_relative
Residual Gate (1, 0.5) Vector-residual_gate_history_a_mean_relative and residual_gate_history_b_mean_relative
Residual Gate (1, 0)-residual_gate_history_a_mean_relative and residual_gate_history_b_mean_relative
Residual Gate (1, 0) Vector-residual_gate_history_a_mean_relative and residual_gate_history_b_mean_relative
Random Learnable Scalar-residual_gate_history_a_mean_relative and residual_gate_history_b_mean_relative
Random Learnable Vector-residual_gate_history_a_mean_relative and residual_gate_history_b_mean_relative
residual_gate_history_a_mean_relative
residual_gate_history_b_mean_relative
Epoch

2.2.5 不同 scheduler 对比实验

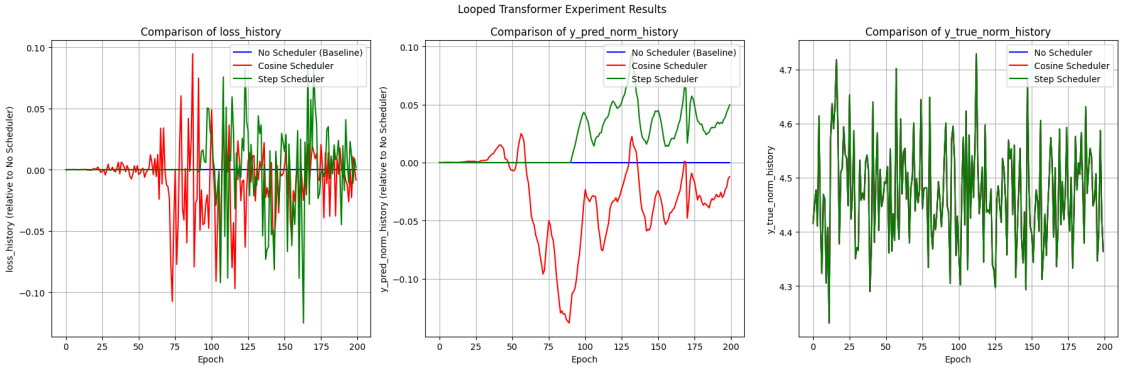
[28]:
```
experiment_table = ExperimentTable(params_groups=[{'experiment_name': 'No␣
↪Scheduler'},
                                                   {'scheduler_type': 'cosine',␣
↪'experiment_name': 'Cosine Scheduler', 'lr_scale': 0},
                                                   {'scheduler_type': 'step',␣
↪'experiment_name': 'Step Scheduler', 'lr_scale': 0.1, 'step_size_scheduler':␣
↪90}],
   manual={'epochs':200, 'lr': 1e-3, 'optimizer_type': 'adamw', 'pe_type':␣
↪['alibi'],
   'wd_adamw': 0.2, 'd_model': 32, 'num_heads': 1, 'max_seq_len': 500,␣
↪'seq_len': 400, 'batch_size': 64, 'd_x': 20,
   'num_blocks': 20, 'num_eff': 15,
   'residual_gate': (1,1), 'residual_gate_type': 'learnable_scalar',␣
↪'gate_lr_ratio': 100,
   'scheduler_type': None, 'lr_scale': 0.01, 'step_size_scheduler': 10,
   'print_every': 50,
   'layer_weight_decay': 0.8, 'seq_weight_decay': 0.9
   })
experiment_table.run(result_lists=[(['loss_history'], 'epoch', 0),␣
↪(['y_pred_norm_history'], 'epoch', 0), (['y_true_norm_history'], 'epoch')])
experiment_table.plot(figure_size=(18, 6))
```

```
No Scheduler initialized in 0.02 seconds. Using device: mps
Cosine Scheduler initialized in 0.01 seconds. Using device: mps
Step Scheduler initialized in 0.01 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.04 GB
Epoch 50/200, Loss: 19.8760, current_blocks: 20
Epoch 100/200, Loss: 21.0343, current_blocks: 20
Epoch 150/200, Loss: 19.6615, current_blocks: 20
Epoch 200/200, Loss: 18.4375, current_blocks: 20
Training completed in 66.28 seconds.
```

[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.02 GB | Metal 驱动分配: 2.13 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.07 GB
Experiment 1/3 completed.

[Experiment 2 训练前] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.08 GB
Epoch 50/200, Loss: 19.8793, current_blocks: 20
Epoch 100/200, Loss: 21.0404, current_blocks: 20
Epoch 150/200, Loss: 19.6585, current_blocks: 20
Epoch 200/200, Loss: 18.4290, current_blocks: 20
Training completed in 64.98 seconds.
[Experiment 2 训练峰值] 显存占用 -> PyTorch 分配: 0.02 GB | Metal 驱动分配: 2.13 GB
[Experiment 2 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.08 GB
Experiment 2/3 completed.

[Experiment 3 训练前] 显存占用 -> PyTorch 分配: 0.01 GB | Metal 驱动分配: 0.09 GB
Epoch 50/200, Loss: 19.8760, current_blocks: 20
Epoch 100/200, Loss: 21.0657, current_blocks: 20
Epoch 150/200, Loss: 19.6840, current_blocks: 20
Epoch 200/200, Loss: 18.4393, current_blocks: 20
Training completed in 65.11 seconds.
[Experiment 3 训练峰值] 显存占用 -> PyTorch 分配: 0.02 GB | Metal 驱动分配: 2.13 GB
[Experiment 3 清理后] 显存占用 -> PyTorch 分配: 0.00 GB | Metal 驱动分配: 0.08 GB
Experiment 3/3 completed.



3 Non-linear Regression Experiments

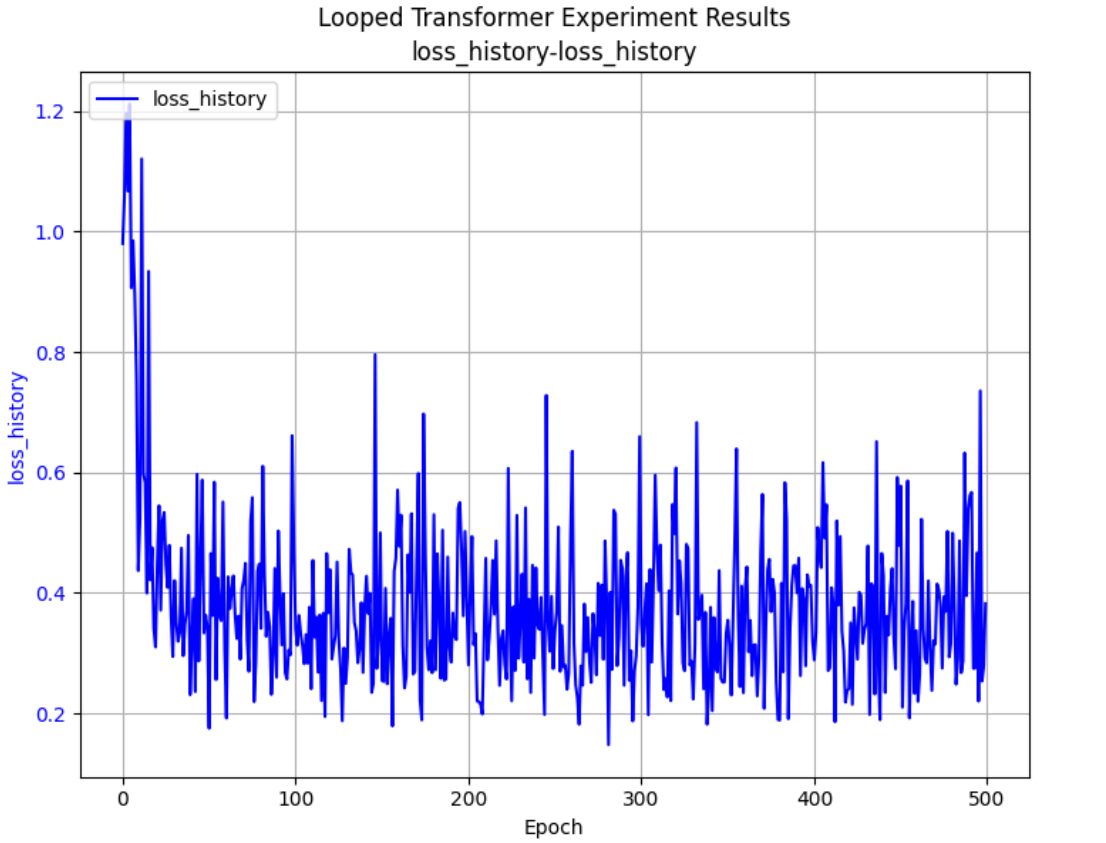
3.1 实验数据观测

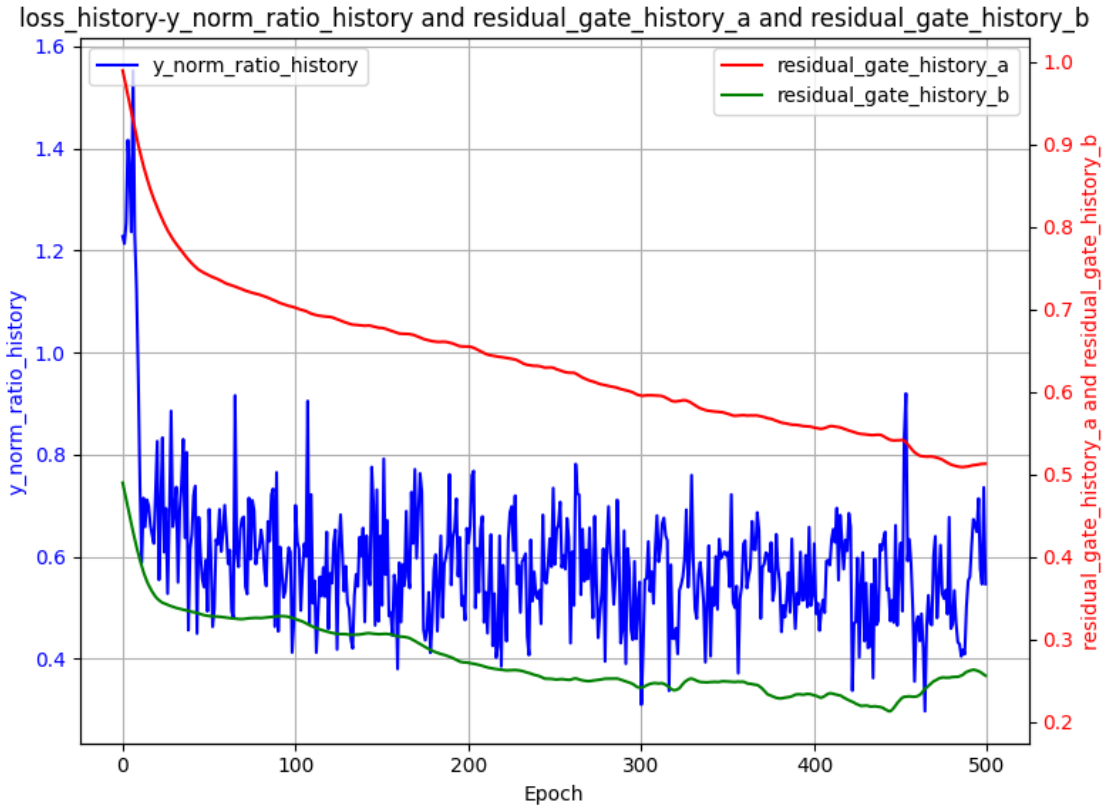
[30]:
```
manual={'data_type': 'nonlinear',
        'function_callable': lambda x: F.relu(x),
        'max_seq_len': 500,
        'scheduled_training': True,
        'pe_type': ['alibi'],
        'residual_gate': (1, 0.5),
        'residual_gate_type': 'learnable_scalar',
        'num_eff': 5,
        'num_blocks': 20,
        'batch_size': 8,
        'num_heads': 2,
        'd_model': 128,
        'seq_len': 400,
        'd_hidden': 8,
        'epochs': 100}
```

[31]:
```
loss_table = ExperimentTable(params_groups=[{'experiment_name': 'loss_history',
                                             'epochs': 500,
                                             'print_every': 100}],␣
 ↪manual=manual)
loss_table.run(result_lists=[(['loss_history'], 'epoch'),␣
 ↪(['y_norm_ratio_history', 'residual_gate_history_a',␣
 ↪'residual_gate_history_b'], 'epoch')])
loss_table.plot(figure_size=(8,12), compare_experiments=False,␣
 ↪subplot_shape=(-1,1))
```

```
loss_history initialized in 0.02 seconds. Using device: mps
[本底显存检测] 显存占用 -> PyTorch 分配: 0.22 GB | Metal 驱动分配: 0.28 GB
[Experiment 1 训练前] 显存占用 -> PyTorch 分配: 0.22 GB | Metal 驱动分配: 0.28 GB
Epoch 100/500, Loss: 0.4739, Norm Ratio: 0.4741, Current Blocks: 20
Epoch 200/500, Loss: 0.3862, Norm Ratio: 0.6557, Current Blocks: 20
Epoch 300/500, Loss: 0.6595, Norm Ratio: 0.5505, Current Blocks: 20
Epoch 400/500, Loss: 0.3205, Norm Ratio: 0.5234, Current Blocks: 20
Epoch 500/500, Loss: 0.3823, Norm Ratio: 0.5465, Current Blocks: 20
```

```
Training completed in 51.75 seconds.
[Experiment 1 训练峰值] 显存占用 -> PyTorch 分配: 0.23 GB | Metal 驱动分配: 0.93 GB
[Experiment 1 清理后] 显存占用 -> PyTorch 分配: 0.22 GB | Metal 驱动分配: 0.29 GB
Experiment 1/1 completed.
```


Looped Transformer Experiment Results
loss_history-loss_history
loss_history
loss_history
Epoch
0
100
200
300
400
500
0.2
0.4
0.6
0.8
1.0
1.2


loss_history-y_norm_ratio_history and residual_gate_history_a and residual_gate_history_b
y_norm_ratio_history
residual_gate_history_a
residual_gate_history_b
y_norm_ratio_history
residual_gate_history_a and residual_gate_history_b
Epoch
0
100
200
300
400
500
0.4
0.6
0.8
1.0
1.2
1.4
1.6
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0

实验数据分析 1. 参数量分析 2. norm_ratio 与 loss 的关系分析 - 对于 Linear+ReLU+Linear，由于 Linear 矩阵归一化后方差为 1，ReLU 又会丢掉一半的激活，因此 y_true 的二阶矩（能量）期望为 $\mathbb{E}[y^2] \approx 0.5$ - 模型在进行 In-Context Learning (ICL) 时，遇到了认知瓶颈。它把复杂的真实信号 y 拆解成了两个部分：$y = y_{learnable} + y_{noise}$ - $y_{learnable}$：模型通过现有的上下文能够学明白的 "低频/线性" 规律。- y_{noise}：由于非线性过于复杂或上下文信息仍显不足，模型无法解开的 "高频/非线性" 残差。- 在 MSE 损失函数的逼迫下，模型最聪明的做法就是精准预测自己能看懂的部分，对看不懂的部分直接输出 0。因此它的预测值 $\hat{y} \approx y_{learnable}$。- 根据 Norm Ratio 倒推 loss：- 观察到 Norm Ratio 在 0.4 附近震荡，即

$$NormRatio = \sqrt{\frac{\mathbb{E}[\hat{y}^2]}{\mathbb{E}[y^2]}} \approx \sqrt{\frac{\mathbb{E}[y^2_{learnable}]}{\mathbb{E}[y^2_{learnable}] + \mathbb{E}[y^2_{noise}]}} \approx 0.4$$

- 代入 $\mathbb{E}[y^2] \approx 0.5$，可以推断出 $\mathbb{E}[y^2_{learnable}] \approx 0.08$，$\mathbb{E}[y^2_{noise}] \approx 0.42$。- 由于 $\hat{y} \to y_{learnable}$，因此最终的 loss 约为

$$\mathbb{E}[(y - \hat{y})^2] = \mathbb{E}[y^2_{noise}] + \mathbb{E}[(y_{learnable} - \hat{y})^2]] \approx 0.42$$

与实际观察到的 loss 值相符。